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Post-selection inference for penalized M-estimators via score thinning

Ronan Perry[1]*, Snigdha Panigrahi[2], Daniela Witten[1,3]

[1] University of Washington, Department of Statistics
[2] University of Michigan, Department of Statistics
[3] University of Washington, Department of Biostatistics

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Abstract

We consider inference for M-estimators after model selection using a sparsity-inducing penalty. While existing methods for this task require bespoke inference procedures, we propose a simpler approach, which relies on two insights: (i) adding and subtracting carefully-constructed noise to a Gaussian random variable with unknown mean and known variance leads to two *independent* Gaussian random variables; and (ii) both the selection event resulting from penalized M-estimation, and the event that a standard (non-selective) confidence interval for an M-estimator covers its target, can be characterized in terms of an approximately normal "score variable". We combine these insights to show that — when the noise is chosen carefully — there is asymptotic independence between the model selected using a noisy penalized M-estimator, and the event that a standard (non-selective) confidence interval on noisy data covers the selected parameter. Therefore, selecting a model via penalized M-estimation (e.g. `glmnet` in `R`) on noisy data, and then conducting *standard* inference on the selected model (e.g. `glm` in `R`) using noisy data, yields valid inference: *no bespoke methods are required*. Our results require independence of the observations, but only weak distributional requirements. We apply the proposed approach to conduct inference on the association between sex and smoking

*Corresponding author: rflperry@uw.edu

in a social network. *Keywords:* Selective inference, post-selection inference, lasso, data thinning

1 Introduction

In classical statistics, we collect data Y in order to conduct inference on a pre-determined target parameter θ_0. However, in practice, a researcher may use the data to select a data-driven parameter $\theta_0(Y)$, and then conduct inference using the same data Y; this is sometimes referred to as *double dipping* [Kriegeskorte et al., 2009]. In general, if we treat $\theta_0(Y)$ *as if it were fixed* and use classical inference techniques, then standard inferential guarantees (such as Type 1 error control of p-values, and coverage of confidence intervals) will not apply [Hotelling, 1940, Cox, 1975, Benjamini et al., 2009, Berk et al., 2013, Fithian et al., 2017, Kuchibhotla et al., 2022, Neufeld et al., 2026].

In recent years, particular interest has focused on inference on models selected via sparsity-penalized M-estimation, such as the lasso [Tibshirani, 1996]. The goal is to develop p-values that control the Type 1 error and confidence intervals that attain the nominal coverage *conditional on the selected model* [Fithian et al., 2017]. We briefly review this line of work in Section 1.1, and then summarize the contributions of our paper in Section 1.2.

1.1 Existing approaches for inference after model selection

First, we review methods that are specifically tailored for inference on a model selected via penalized M-estimation. The conditional selective inference proposal of Lee et al. [2016] provides valid inference on linear contrasts of the mean of a Gaussian response vector when these contrasts were selected by the lasso; this is accomplished by characterizing the lasso selection event as a set of linear inequalities, and then deriving the distribution of the estimator conditional on the lasso selection event. Tibshirani et al. [2018] and Tian and Taylor [2017] extend these results to

provide asymptotic guarantees without distributional assumptions on the response vector, and Tian and Taylor [2017] allow the number of features p to grow with the sample size n. These methods provide valid inference conditional on the event that a specific model was selected via the lasso using all of the data. However, this framework requires analytical characterization of the selection event for each model selection procedure; furthermore, it lacks power near the boundary of the selection event [Kivaranovic and Leeb, 2021].

To improve power, randomized conditional selective inference methods introduce noise into the selection process. In their pioneering work, Tian and Taylor [2018] prove a distribution-free selective central limit theorem which they use to derive a selective density by conditioning on having selected a model using an L_1-penalty. However, this selective density must be approximated via sampling. In a linear Gaussian model, Panigrahi et al. [2024] derive a numerically-tractable closed-form distribution for inference after the randomized lasso, while Bakshi et al. [2024] extend this to L_1-penalized M-estimators. In the context of group-lasso-penalized M-estimators, Huang et al. [2024] derive a selective density which can be numerically approximated. Once again, these approaches require the derivation of a bespoke method for inference for each model selection procedure.

We now turn to approaches that split the data into two independent sets, so that the first can be used to select a model and the second to conduct inference. A major advantage of such approaches over the aforementioned (randomized) conditional selective inference approaches is that no bespoke methods for inference are required, i.e., valid inference can be obtained using standard inference methods and software, e.g., `glm` and `glmnet` in `R` [Friedman et al., 2010]. Provided that multiple independent and identically-distributed random variables are available, sample splitting [Cox, 1975] is a straightforward option. However, if the observations are not independent then sample splitting is not applicable. Even if they are independent, sample splitting can be unattractive if they are not identically distributed, or if the sample size is very small [Rasines and Young, 2023, Dharamshi et al., 2025]. Recent proposals have moved beyond sample splitting: Robins and van der

Vaart [2006] show that a single Gaussian random variable Y with unknown mean and known variance can be split into two independent Gaussians $Y^{(1)}$ and $Y^{(2)}$; this fact was subsequently applied to construct confidence intervals [Tian and Taylor, 2018, Rasines and Young, 2023] and to estimate prediction error [Tian, 2020, Oliveira et al., 2024] after model selection. Leiner et al. [2025] further considered a similar decomposition in the Poisson case, while Neufeld et al. [2024] and Dharamshi et al. [2025] generalize this idea to a much broader set of distributions and refer to this generalization as "data thinning".

The distributional assumptions required for data thinning are restrictive, and recent work seeks to loosen them in exchange for asymptotic (rather than finite-sample) guarantees for i.i.d. data. Rasines and Young [2023] show that it is possible to thin the data Y as though it were Gaussian and still obtain asymptotically valid inference on linear contrasts of $\mathbb{E}[Y^{(2)}]$ when the selection event is that $Y^{(1)}$ lives in a convex set. Lei [2025] provides asymptotically valid inference on linear contrasts of $\mathbb{E}[Y^{(2)}]$ when the selection event is any function of the contrasts $X^\top Y^{(1)}$ under certain conditions, e.g., (i) when the selection event is that $X^\top Y^{(1)}$ lives in a convex set, or (ii) a total-variation bound requiring continuous errors $Y - \mathbb{E}[Y]$; notably, these conditions do not hold in the case of logistic regression. Separately, Panigrahi et al. [2024] and Fry et al. [2025] demonstrate how to asymptotically thin an estimator, e.g., the OLS estimator, using any available bootstrap procedure with a CLT. However, this does not enable inference on a selected model; instead, it enables inference on a selected parameter in a *fixed* model.

Finally, it bears mention that a line of work by Meinshausen et al. [2009], Zhang and Zhang [2014], Javanmard and Montanari [2014], Sengupta and Janson [2025] uses the lasso as a tool to conduct inference on an underlying sparse, high-dimensional linear model. Thus, their target parameter is *not* a function of the data.

1.2 Contributions of this paper

In this paper, we consider inference on a parameter in a model selected via penalized M-estimation, e.g., the lasso. We do not make distributional assumptions, nor do we assume identically distributed observations. Our developments rely upon a key insight: namely, that both selection and inference hinge upon the same approximately normal *score variable*, $\hat{Z}_n$.

In greater detail, our insights are as follows:

Insight 1: Since $\hat{Z}_n$ is approximately normal, it can be decomposed into two pieces, $\hat{Z}_n^{(1)}$ and $\hat{Z}_n^{(2)}$, which are approximately normal and independent, up to appropriately vanishing remainders.

Insight 2: The event that a particular model is selected via "noisy" penalized M-estimation is equivalent to an event involving $\hat{Z}_n^{(1)}$. Furthermore, the event that a standard (non-selective) confidence interval centered around a "noisy" M-estimator covers its target is equivalent to an event involving $\hat{Z}_n^{(2)}$.

Insight 3: Combining Insights 1 and 2, if we select a model using a "noisy" penalized M-estimator, then a standard confidence interval centered around a "noisy" M-estimator attains the nominal coverage, conditional on the selection event.

Thus, *we can conduct valid post-selection inference for models selected via (noisy) penalized M-estimation using standard inference pipelines.* This is in marked contrast to the methods discussed in Section 1.1, which require bespoke inference techniques [Lee et al., 2016, Fithian et al., 2017, Tian and Taylor, 2017, Tibshirani et al., 2018, Tian and Taylor, 2018, Panigrahi et al., 2024, Huang et al., 2024, Leiner et al., 2025], stringent distributional assumptions [Neufeld et al., 2024, Dharamshi et al., 2025], or estimands that are limited to linear contrasts of i.i.d. outcomes [Rasines and Young, 2023, Lei, 2025].

The rest of this paper is organized as follows. In Section 2, we formalize Insight 1. In Section 3, we formalize Insights 2 and 3. In Section 4, we show that our theory can be applied to conduct inference on parameters in generalized linear models selected via L_1 penalization. In Section 5, we examine the validity of our method in

simulations and apply it to conduct inference after principal components network regression on the *Teenage Friends and Lifestyle Study* dataset. Section 6 concludes with a discussion.

2 Valid post-selection inference via a score variable

We review the M-estimation setting in Section 2.1, present the score variable $\hat{Z}_n$ in Section 2.2, and discuss the construction and properties of $\hat{Z}_n^{(1)}$ and $\hat{Z}_n^{(2)}$ in Section 2.3.

2.1 M-estimation

Suppose that $Y_1, \ldots, Y_n \overset{\text{ind.}}{\sim} F_n$ form a triangular array of independent random variables $Y_i \in \mathcal{Y} \subseteq \mathbb{R}^d$ for some $d > 0$, keeping the dependence of the distribution of Y_i on n implicit. Let $\ell_\theta : \mathcal{Y} \mapsto \mathbb{R}$ be a differentiable loss function parameterized by $\theta \in \mathbb{R}^p$. We will implicitly allow this loss function to depend on the index i, e.g., in the case of fixed-X regression where fixed covariates X_i may be available.

Let $E \subseteq [p]$ denote the indices of any non-empty subset of the parameters. Without loss of generality, we assume that $E = \{1, 2, \ldots, |E|\}$. Our goal is to conduct inference on the population estimand corresponding to the model with features indexed by E,

$$\theta_n^{E,*} = \underset{\theta \in \mathbb{R}^{|E|}}{\arg\min} \left\{ \frac{1}{n} \sum_{i=1}^{n} \mathbb{E}\left[\ell_{(\theta, 0_{p-|E|})}(Y_i) \right] \right\} = \underset{\theta \in \mathbb{R}^{|E|}}{\arg\min} \left\{ P_n \mathbb{E} \ell_{(\theta, 0_{p-|E|})} \right\}, \tag{2.1}$$

where the last equality makes use of shorthand that will appear throughout this paper: namely, $P_n f := \frac{1}{n} \sum_{i=1}^{n} f(Y_i)$, for any function f defined over $\mathcal{Y}$.

For fixed model E, level $\alpha \in (0, 1)$, and contrast $\xi \in \mathbb{R}^{|E|}$, consider the function $\text{CI}_n^{\xi,\alpha} : \left(\mathbb{R}^{|E|} \times \mathbb{S}_{|E|}^{++} \right) \mapsto \mathbb{R}^2$, where $\mathbb{S}_{|E|}^{++}$ denotes the space of $|E| \times |E|$ positive

definite matrices. For any $\theta^E \in \mathbb{R}^{|E|}$ and $S^E \in \mathbb{S}^{++}_{|E|}$, we define

$$\mathrm{CI}_n^{\xi,\alpha}\left(\theta^E, S^E\right) := \left[\xi^\top \theta^E \pm n^{-1/2} z_{1-\alpha/2}\left(\xi^\top S^E \xi\right)^{1/2}\right], \tag{2.2}$$

where $z_{1-\alpha/2}$ is the $(1-\alpha/2)$-quantile of a standard normal distribution. Then, for the M-estimator $\tilde{\theta}_n^E \in \mathbb{R}^{|E|}$ based on the model E and defined as

$$\tilde{\theta}_n^E := \underset{\theta \in \mathbb{R}^{|E|}}{\arg\min}\left\{P_n \ell_{\left(\theta, 0_{p-|E|}\right)}\right\}, \tag{2.3}$$

and for the sandwich variance estimator $\tilde{S}_n^E := \left[P_n \ddot{\ell}_{\tilde{\theta}_n^E}\right]^{-1} P_n\left[\dot{\ell}_{\tilde{\theta}_n^E}\dot{\ell}^\top_{\tilde{\theta}_n^E}\right]\left[P_n \ddot{\ell}_{\tilde{\theta}_n^E}\right]^{-1} \in \mathbb{S}^{++}_{|E|}$, it holds that $\mathrm{CI}_n^{\xi,\alpha}\left(\tilde{\theta}_n^E, \tilde{S}_n^E\right)$ is a valid confidence interval under typical conditions [van der Vaart, 1998, Section 5.2.1], in the sense that it attains the nominal (asymptotic) coverage, i.e.,

$$\lim_{n\to\infty} \Pr\left(\xi^\top \theta_n^{E,*} \in \mathrm{CI}_n^{\xi,\alpha}\left(\tilde{\theta}_n^E, \tilde{S}_n^E\right)\right) \geq 1-\alpha. \tag{2.4}$$

Now, suppose instead that E is random: for instance, it may be the support of a penalized M-estimator, i.e., $E := \mathrm{supp}\left(\tilde{\theta}_n^{(\lambda)}\right)$, where

$$\tilde{\theta}_n^{(\lambda)} \in \underset{\theta \in \mathbb{R}^p}{\arg\min}\left\{P_n \ell_\theta + n^{-1/2}\rho_\lambda(\theta)\right\}, \tag{2.5}$$

and $\rho_\lambda : \mathbb{R}^p \mapsto \mathbb{R}$ is a non-differentiable penalty parameterized by a fixed tuning parameter λ. For example, using the L_1-penalty $\rho_\lambda(\theta) := \lambda \|\theta\|_1$ and squared-error loss $\ell_\theta(X_i, Y_i) := \frac{1}{2}(Y_i - X_i^\top \theta)^2$, (2.5) is the popular lasso estimator [Tibshirani, 1996]. When E is random, the interval $\mathrm{CI}_n^{\xi,\alpha}\left(\tilde{\theta}_n^E, \tilde{S}_n^E\right)$ is no longer guaranteed to achieve the nominal coverage, i.e., (2.4) does not hold. Our goal in this paper is to construct estimators $\hat{\theta}_n^E$ and $\hat{S}_n^E$ such that the interval $\mathrm{CI}_n^{\xi,\alpha}\left(\hat{\theta}_n^E, \hat{S}_n^E\right)$ attains the nominal coverage, *conditional* on the fact that E is the support of some penalized M-estimator $\hat{\theta}_n^{(\lambda)}$ that remains to be specified. That is, we wish for the conditional coverage guarantee

$$\lim_{n\to\infty} \Pr\left(\xi^\top \theta_n^{E,*} \in \mathrm{CI}_n^{\xi,\alpha}\left(\hat{\theta}_n^E, \hat{S}_n^E\right) \,\middle|\, \mathrm{supp}\left(\hat{\theta}_n^{(\lambda)}\right) = E\right) \geq 1-\alpha. \tag{2.6}$$

2.2 The score variable

It should not be surprising that since (2.3) and (2.5) are solutions to a minimization problem, the gradient of the loss function ℓ_θ with respect to θ plays an important role. In fact, this gradient defines the score variable. Recalling the definition of $\theta_n^{E,*}$ in (2.1), we define

$$\check{\theta}_n^{E,*} := (\theta_n^{E,*}, 0_{|E^c|}), \tag{2.7}$$

the expanded p-dimensional vector. For any θ, define the gradient

$$\dot{\ell}_\theta(Y_i) := \left[\frac{\partial}{\partial\theta'_1}\ell_{\theta'}(Y_i), \ldots, \frac{\partial}{\partial\theta'_p}\ell_{\theta'}(Y_i)\right]^\top \Bigg|_{\theta'=\theta} \in \mathbb{R}^p. \tag{2.8}$$

Under standard conditions for M-estimation (in which E is fixed), the E-indices of the score variable

$$\hat{Z}_n := \sqrt{n} P_n \dot{\ell}_{\check{\theta}_n^{E,*}} \tag{2.9}$$

are asymptotically normal; this property forms the basis for asymptotically valid inference. However, here our interest lies in asymptotically valid inference when the set E arises from model selection. In this setting, we will show that $\hat{Z}_n$ is notable in that it contains all of the information needed for both selection *and* inference, up to a vanishing remainder.

2.3 Approximate normality of the thinned score variable

We will now borrow ideas from the literature on data thinning [Rasines and Young, 2023, Neufeld et al., 2024, Dharamshi et al., 2025] to decompose the score variable $\hat{Z}_n$ in (2.9) into two components, which we will then show are approximately normally distributed, in the sense of a Berry-Esseen-type bound [Raič, 2019], and approximately conditionally independent, in a sense that we will specify. This formalizes Insight 1 in Section 1.2.

Using the notation defined in (2.8), we require the following conditions. Let $\mu_{n,i}^{E,*} := \mathbb{E}\left[\dot{\ell}_{\check{\theta}_n^{E,*}}(Y_i)\right]$ and $\mu_n^{E,*} := \frac{1}{n}\sum_{i=1}^n \mu_{n,i}^{E,*}$.

Condition 1.

(i) $\Sigma_n := \frac{1}{n}\sum_{i=1}^{n} \mathrm{Var}\left(\dot{\ell}_{\check{\theta}_n^{E,*}}\right)$ is invertible.

(ii) $P_n \mathbb{E}\left\|\dot{\ell}_{\check{\theta}_n^{E,*}} - \mu_{n,i}^{E,*}\right\|_2^3$ is uniformly bounded above.

Under Condition 1, it can be shown that $\hat{Z}_n$ is approximately $\mathrm{N}\left(\sqrt{n}\mu_n^{E,*}, \Sigma_n\right)$ [Raič, 2019]. We now apply Gaussian data thinning to $\hat{Z}_n$ as though it were exactly normally distributed. That is, for some $\hat{\Sigma}_n \in \mathbb{R}^{p\times p}$, we define $\hat{W}_n \sim \mathrm{N}\left(0, \hat{\Sigma}_n\right)$ and

$$\hat{Z}_n^{(1)} := \hat{Z}_n + \gamma \hat{W}_n \quad \text{and} \quad \hat{Z}_n^{(2)} := \hat{Z}_n - \frac{1}{\gamma}\hat{W}_n. \tag{2.10}$$

The tuning parameter γ determines the amount of noise added to $\hat{Z}_n$ to construct $\hat{Z}_n^{(1)}$ versus $\hat{Z}_n^{(2)}$. When $\gamma = 1$, the same amount of noise is added to each, and the procedure evokes splitting n samples into two equal pieces.

The following Berry-Esseen-type bound, using results from Raič [2019], establishes that $\left(\hat{Z}_n^{(1)}, \hat{Z}_n^{(2)}\right)$ are approximately distributed as $\left(Z_n^{(1)}, Z_n^{(2)}\right)$, where

$$Z_n^{(1)} \sim \mathrm{N}\left(\sqrt{n}\mu_n^{E,*}, \left(1+\gamma^2\right)\Sigma_n\right) \quad \text{and} \quad Z_n^{(2)} \sim \mathrm{N}\left(\sqrt{n}\mu_n^{E,*}, \left(1+\tfrac{1}{\gamma^2}\right)\Sigma_n\right) \tag{2.11}$$

are independent random variables. Let $\lambda_{\min}(\cdot)$ denote the minimum eigenvalue of its argument, a positive semi-definite matrix.

Lemma 1. *Suppose that $C \subseteq \mathbb{R}^{2p}$ is the union of K measurable and almost surely disjoint convex sets. Under Condition 1, for any $\Delta \geq 0$,*

$$\begin{aligned}
&\left|\Pr\left(\left(\hat{Z}_n^{(1)}, \hat{Z}_n^{(2)}\right) \in C\right) - \Pr\left(\left(Z_n^{(1)}, Z_n^{(2)}\right) \in C\right)\right| \\
&\leq \Pr\left(\left\|\hat{\Sigma}_n - \Sigma_n\right\|_F > \Delta\lambda_{\min}\left(\Sigma_n\right)\right) + O\left(Kp^{1/4}n^{-1/2}\lambda_{\min}^{-3/2}\left(\Sigma_n\right) + \Delta\right).
\end{aligned} \tag{2.12}$$

When $\hat{\Sigma}_n = \Sigma_n$, i.e., the covariance $\hat{\Sigma}_n$ used to construct $\hat{Z}_n^{(1)}$ and $\hat{Z}_n^{(2)}$ in (2.10) exactly equals Σ_n defined in Condition 1, then we can set $\Delta = 0$ to obtain the classical Berry-Esseen-type rate. We now establish a conditional Berry-Esseen-type result.

Lemma 2. *Let A_n and B_n be sets such that $A_n \times B_n$ is the union of K measurable*

and almost surely disjoint convex sets, and let r_n^A *and* r_n^B *be random vectors in* $\mathbb{R}^p$. *For any* $\Delta \geq 0$, *and* $\varepsilon > 0$ *smaller than the smallest inradius of those* K *sets* (B.18), *define the rates*

$$E_1 := p^{1/4} n^{-1/2} \lambda_{\min}^{-3/2}(\Sigma_n) + \Delta + \Pr\left(\left\|\hat{\Sigma}_n - \Sigma_n\right\|_F > \Delta \lambda_{\min}(\Sigma_n)\right),$$
$$E_2 := O\left(E_1 + p^{1/2}\varepsilon\left(\lambda_{\min}^{1/2}(\Sigma_n) + \lambda_{\min}^{-1/2}(\Sigma_n)\right)\right) + \Pr\left(\left\|r_n^A\right\|_2 > \tfrac{\varepsilon}{2}\right) + \Pr\left(\left\|r_n^B\right\|_2 > \tfrac{\varepsilon}{2}\right),$$
$$E_3 := \Pr\left(Z_n^{(1)} \in A_n^{-\varepsilon}\right) - O(E_1) - \Pr\left(\left\|r_n^A\right\|_2 > \varepsilon\right),$$

where $A_n^{-\varepsilon}$ *is the* ε*-shrinkage of the set* A_n, *as defined in* (B.15). *Under Condition 1,*

$$\left|\Pr\left(\hat{Z}_n^{(2)} + r_n^B \in B_n \middle| \hat{Z}_n^{(1)} + r_n^A \in A_n\right) - \Pr\left(Z_n^{(2)} \in B_n\right)\right| \leq \frac{KE_2}{\max\{E_3, KE_2\}}. \tag{2.13}$$

In the next section, we will show that Lemma 2 enables valid post-selection inference on a model selected via penalized M-estimation.

3 Asymptotically valid post-selection inference

In Sections 3.1 and 3.2, we apply Lemma 2 to establish two approaches for valid conditional selective inference in the asymptotic regime where p is fixed and n grows.

3.1 Post-selection inference by adding a noisy penalty

Suppose that $\gamma > 0$ and $\hat{W}_n \sim \mathrm{N}(0, \hat{\Sigma}_n)$ for any covariance matrix $\hat{\Sigma}_n$. We consider inference on the selected model $\operatorname{supp}\left(\hat{\theta}_n^{(\lambda)}\right)$, where $\hat{\theta}_n^{(\lambda)}$ is the solution to a penalized M-estimation problem, i.e.

$$\hat{\theta}_n^{(\lambda)} = \underset{\theta \in \mathbb{R}^p}{\arg\min}\left\{P_n \ell_\theta + \gamma n^{-1/2} \hat{W}_n^\top \theta + n^{-1/2} \rho_\lambda(\theta)\right\}. \tag{3.1}$$

Note that (3.1) is a variant of (2.5), with the addition of the noisy penalty term $\gamma n^{-1/2} \hat{W}_n^\top \theta$; it resembles randomized estimators seen in the conditional selective

inference literature [Tian and Taylor, 2018, Panigrahi et al., 2024, Huang et al., 2024].

We will further consider the randomized M-estimator on the sub-model with features in the index set $E \in \{1, \ldots, p\}$, where $\hat{W}_{n,E}$ is the $|E|$-dimensional vector obtained by subsetting the elements of $\hat{W}_n$ to those indexed by E:

$$\hat{\theta}_n^E := \underset{\theta \in \mathbb{R}^{|E|}}{\arg\min} \left\{ P_n \ell_{(\theta, 0_{p-|E|})} - \tfrac{1}{\gamma} n^{-1/2} \hat{W}_{n,E}^\top \theta \right\}. \tag{3.2}$$

We note that (3.2) is a variant of (2.3) that incorporates the added noisy penalty term $-\frac{1}{\gamma} n^{-1/2} \hat{W}_n^\top \theta$. As in $\check{\theta}_n^{E,*}$, we define the expanded p-dimensional minimizer

$$\check{\theta}_n^E := \left(\hat{\theta}_n^E, 0_{|E^c|} \right). \tag{3.3}$$

We require these estimators to be unique almost surely.

Condition 2. The solutions to (3.1) and (3.2) are unique almost surely.

Furthermore, we require the existence of a valid *fixed model* confidence interval (2.2) around our point estimator (3.2).

Condition 3. There exists an estimator $\hat{S}_n^E \in \mathbb{S}_{|E|}^{++}$ satisfying

$$\lim_{n \to \infty} \Pr \left(\xi^\top \theta_n^{E,*} \in \mathrm{CI}_n^{\xi,\alpha} \left(\hat{\theta}_n^E, \hat{S}_n^E \right) \right) \geq 1 - \alpha.$$

Before delving further into technical details, we outline the key intuition of this section. Recall the construction of $\hat{Z}_n^{(1)}$ and $\hat{Z}_n^{(2)}$ from (2.10). First, in Lemma 4, we will show that for any index set $E \subseteq \{1, \ldots, p\}$, the event $\left\{ \mathrm{supp} \left(\hat{\theta}_n^{(\lambda)} \right) = E \right\}$ equals the event that $\hat{Z}_n^{(1)}$, plus a small remainder term, belongs to some union of convex sets. Then, in Lemma 5, we will show that the event $\left\{ \xi^\top \theta_n^{E,*} \in \mathrm{CI}_n^{\xi,\alpha} \left(\hat{\theta}_n^E, \hat{S}_n^E \right) \right\}$ equals the event that $\hat{Z}_n^{(2)}$, plus a small remainder term, belongs to some convex set, where $\theta_n^{E,*}$ was defined in (2.1). Together, Lemmas 4 and 5 will formalize Insight 2 from Section 1.2. Combining these results with Lemma 2 will enable us to show, in Theorem 6, that the event $\left\{ \xi^\top \theta_n^{E,*} \in \mathrm{CI}_n^{\xi,\alpha} \left(\hat{\theta}_n^E, \hat{S}_n^E \right) \right\}$ holds with probability

$(1-\alpha)$ asymptotically, conditional on the selection event $\left\{\text{supp}\left(\hat{\theta}_n^{(\lambda)}\right) = E\right\}$; this formalizes Insight 3.

We first require that our loss function satisfies certain typical M-estimation properties, namely that it converges uniformly across the parameter space and can be well-approximated by a second-order Taylor expansion. This approximation is well-studied in the M-estimation literature and will be discussed in further detail in Section 4 in the context of generalized linear models.

Condition 4. Our estimators and loss function admit the following Taylor expansions centered at $\check{\theta}_n^{E,*}$ defined in (2.7):

$$P_n \dot{\ell}_{\hat{\theta}_n^{(\lambda)}} = P_n \dot{\ell}_{\check{\theta}_n^{E,*}} + H_n \left(\hat{\theta}_n^{(\lambda)} - \check{\theta}_n^{E,*}\right) + o_p\left(n^{-1/2}\right)$$

and

$$P_n \dot{\ell}_{\check{\theta}_n^{E}} = P_n \dot{\ell}_{\check{\theta}_n^{E,*}} + H_n \left(\check{\theta}_n^{E} - \check{\theta}_n^{E,*}\right) + o_p\left(n^{-1/2}\right),$$

for some deterministic matrix $H_n \in \mathbb{S}_p^{++}$, and where $\dot{\ell}_\theta$ was defined in (2.8).

Furthermore, we assume that the following population quantities converge asymptotically to non-degenerate quantities.

Assumption 1. Recalling the definitions of $\theta_n^{E,*}$ in (2.1), Σ_n in Condition 1, and H_n in Condition 4, the following limits exist:

(i) $\sqrt{n}\theta_n^{E,*} \to \theta^{E,*}$ for some fixed and bounded $\theta^{E,*}$,

(ii) $\Sigma_n \to \Sigma$ for some fixed $\Sigma \in \mathbb{S}_p^{++}$,

(iii) $H_n \to H$ for some fixed $H \in \mathbb{S}_p^{++}$.

In our theoretical development, Assumption 1(i) will guarantee that the probability of selecting the model E is asymptotically bounded above zero. Similar local alternative-type assumptions are made in related work [Tian and Taylor, 2018, Tibshirani et al., 2018]. In contrast, Lei [2025] guarantee only unconditional selective coverage and thus do not require such an assumption, while Rasines and Young [2023] make the weaker but less verifiable assumption that the probability of selection decays to zero slow enough.

We also place a very mild set of conditions on the penalty function.

Condition 5. The penalty $\rho_\lambda(\theta)$ satisfies the following properties:

(i) The subgradients are uniformly bounded, i.e., $\sup_{\theta\in\mathbb{R}^p}\sup_{\eta\in\partial\rho(\theta)}\|\eta\|_2<\infty$;

(ii) For any $H\in\mathbb{S}_p^{++}$,

$$\{H\theta+\eta:\operatorname{supp}(\theta)=E,\eta\in\partial\rho_\lambda(\theta)\}$$

can be written as the union of finitely-many disjoint convex sets with non-empty interiors.

Condition 5 is satisfied by the popular L_1 penalty. While a similar result is used in the foundational work of Lee et al. [2016], where a linear model with Gaussian errors is assumed, we emphasize that Condition 5 involves only the penalty term $\rho_\lambda(\theta)$ and places no assumptions on the distribution of the observations.

Proposition 3. *For $\lambda\in(0,\infty)$, the lasso penalty $\rho_\lambda(\theta)=\lambda\|\theta\|_1$ satisfies Condition 5.*

Finally, we require that $\hat{\Sigma}_n$ is an appropriate estimator for Σ_n.

Condition 6. $\hat{\Sigma}_n$ satisfies $\mathbb{E}\left[\left\|\hat{\Sigma}_n-\Sigma_n\right\|_F\right]=o(1)$.

Recalling the definitions of $\hat{Z}_n^{(1)}$ and $\hat{Z}_n^{(2)}$ in (2.10), we are now ready to present Lemmas 4 and 5.

Lemma 4. *Under Assumption 1 and Conditions 2–6, the event $\left\{\operatorname{supp}\left(\hat{\theta}_n^{(\lambda)}\right)=E\right\}$ equals the event $\left\{\hat{Z}_n^{(1)}+r_n^A\left(\hat{\theta}_n^{(\lambda)}\right)\in A_n^E\right\}$ for some $r_n^A\left(\hat{\theta}_n^{(\lambda)}\right)$ such that $r_n^A\left(\hat{\theta}_n^{(\lambda)}\right)=o_p(1)$, and for some A_n^E that is the union of finitely many measurable disjoint convex sets.*

Lemma 5. *Under Assumption 1 and Conditions 2, 4, and 6, the event $\left\{\xi^\top\theta_n^{E,*}\in\mathrm{CI}_n^{\xi,\alpha}\left(\hat{\theta}_n^E,\hat{S}_n^E\right)\right\}$ equals the event $\left\{\hat{Z}_n^{(2)}+r_n^B\left(\hat{\theta}_n^E\right)\in B_n^E\right\}$ for some $r_n^B\left(\hat{\theta}_n^E\right)$ such that $r_n^B\left(\hat{\theta}_n^E\right)=o_p(1)$, and for some measurable convex set B_n^E.*

Lemmas 4 and 5 formalize Insight 2 in Section 1.2 by establishing that both the selection and inference events are related to thinned versions of the score variable

$\hat{Z}_n$. We now combine these lemmas with the conditional Berry-Esseen results in Lemma 2 to formalize Insight 3.

Theorem 6 (Post-selection inference via a noisy penalty)**.** *Under Assumption 1 and Conditions 1–6,*

$$\lim_{n\to\infty} \Pr\left(\xi^\top \theta_n^{E,*} \in \mathrm{CI}_n^{\xi,\alpha}\left(\hat{\theta}_n^E, \hat{S}_n^E\right) \middle| \operatorname{supp}\left(\hat{\theta}_n^{(\lambda)}\right) = E\right) \geq 1-\alpha.$$

Theorem 6 tells us that if we select a model E using the randomized penalized M-estimator $\hat{\theta}_n^{(\lambda)}$ in (3.1), then we can conduct inference on the parameter $\theta_n^{E,*}$ corresponding to this selected model by simply constructing a standard confidence interval of the form (2.2) that is centered around the randomized M-estimator $\hat{\theta}_n^E$ in (3.2).

From a computational perspective, this amounts to an astoundingly simple strategy for post-selection inference for penalized M-estimators: valid inference in the model selected by the randomized penalized M-estimator $\hat{\theta}_n^{(\lambda)}$ in (3.1) simply requires software to solve the optimization problem (3.1) (for both $\lambda > 0$ and $\lambda = 0$), e.g., using `adelie` [Yang and Hastie, 2024]. This is in marked contrast to previous proposals from the selective inference literature such as Tibshirani et al. [2018] and Huang et al. [2024], which also consider inference on the model given by the support of $\hat{\theta}_n^{(\lambda)}$ but require bespoke procedures for downstream inference.

3.2 Post-selection inference by noising the samples

The very simple strategy for post-selection inference for penalized M-estimators suggested by Theorem 6 requires the analyst to solve the randomized penalized M-estimation problem given in (3.1). In this section, we will present a corollary of Theorem 6, which reveals that as long as the loss function takes an appropriate form, we can conduct post-selection inference for penalized M-estimators *without* needing to solve (3.1) at all. The key is to add noise to the data Y rather than to the penalized M-estimator optimization problem; this mirrors (but substantially generalizes) the approaches taken by Rasines and Young [2023] and Lei [2025], and

provides an asymptotic and distribution-free version of the finite-sample Gaussian guarantees of Rasines and Young [2023].

Recall from Section 2.1 our random data $Y_i \in \mathbb{R}^d$. Let $\bar{\Sigma}_{n,i} \in \mathbb{R}^{d\times d}$ be an estimate of the covariance $\mathrm{Var}(Y_i)$; in many settings, we will have a homogeneous covariance estimate $\bar{\Sigma}_{n,i} = \bar{\Sigma}_n$, although we write our results here more generally to accommodate arbitrarily heterogeneous covariances. We construct auxiliary noise $\bar{W}_{n,i} \sim \mathrm{N}(0, \bar{\Sigma}_{n,i})$, and define

$$\bar{\theta}_n^{(\lambda)} = \underset{\theta\in\mathbb{R}^p}{\arg\min} \left\{ \frac{1}{n}\sum_{i=1}^n \ell_\theta\left(Y_i + \gamma\bar{W}_{n,i}\right) + n^{-1/2}\rho_\lambda(\theta) \right\} \tag{3.4}$$

and

$$\bar{\theta}_n^{E} := \underset{\theta\in\mathbb{R}^{|E|}}{\arg\min} \left\{ \frac{1}{n}\sum_{i=1}^n \ell_{(\theta,0_{p-|E|})}\left(Y_i - \tfrac{1}{\gamma}\bar{W}_{n,i}\right) \right\}. \tag{3.5}$$

We require the following condition.

Condition 7. The loss function $\ell_\theta(Y_i) = \theta^\top A_i Y_i + a_{\theta,i}$ for some $A_i \in \mathbb{R}^{p\times d}$ and $a_{\theta,i} \in \mathbb{R}$.

Under Condition 7, it is clear that

$$\ell_\theta(Y_i+\gamma\bar{W}_{n,i}) = \ell_\theta(Y_i)+\gamma\theta^\top A_i\bar{W}_{n,i} \quad \text{and} \quad \ell_\theta\left(Y_i - \tfrac{1}{\gamma}\bar{W}_{n,i}\right) = \ell_\theta(Y_i) - \tfrac{1}{\gamma}\theta^\top A_i\bar{W}_{n,i},$$

where $A_i\bar{W}_{n,i} \sim \mathrm{N}\left(0, A_i\bar{\Sigma}_{n,i}A_i^\top\right)$. Thus, for $\hat{W}_n = n^{-1/2}\sum_{i=1}^n A_i\bar{W}_{n,i}$, where $\hat{W}_n$ is the noise used to estimate $\hat{\theta}_n^{(\lambda)}$ in (3.1) and $\hat{\theta}_n^E$ in (3.2), the objective function in (3.1) is equivalent to that of (3.4), and (3.2) is equivalent to (3.5). This implies that Theorem 6 holds.

Corollary 7 (Valid post-selection inference via a noisy response)**.** *Under Condition 7, if* $\hat{W}_n = n^{-1/2}\sum_{i=1}^n A_i\bar{W}_{n,i}$ *then* $\bar{\theta}_n^{(\lambda)} = \hat{\theta}_n^{(\lambda)}$ *and* $\bar{\theta}_n^E = \hat{\theta}_n^E$*. Therefore, under Assumption 1 and Conditions 1–6,*

$$\lim_{n\to\infty} \Pr\left(\xi^\top\theta_n^{E,*} \in \mathrm{CI}_n^{\xi,\alpha}\left(\bar{\theta}_n^E, \hat{S}_n^E\right) \middle| \operatorname{supp}\left(\bar{\theta}_n^{(\lambda)}\right) = E\right) \geq 1-\alpha.$$

Thus, from a computational perspective, valid post-selection inference on a

model selected via penalized M-estimation requires only software to compute a penalized M-estimator, provided that the loss function is of an appropriate form. We will see that this holds for a variety of common M-estimators such as generalized linear models with canonical link functions.

4 Application to generalized linear models

We now apply our results from previous sections to the setting of a fixed-X generalized linear model (GLM). Even in this very simple setting, existing selective inference tools are restrictive. For instance, consider logistic regression. Since the outcomes are Bernoulli, data thinning is not applicable [Dharamshi et al., 2025], and the asymptotic results of Lei [2025] require outcomes with continuous errors. Data fission [Leiner et al., 2025] can be applied in principle, using insights from Neufeld et al. [2025], but challenges remain: for instance, (i) when the logistic regression model does not hold, the parameter targeted by data fission typically does not coincide with the true parameter of interest; and (ii) it is not clear how to target a particular allocation of Fisher information between the selection and inference steps. The asymptotic results of Tian and Taylor [2018] are applicable, but require complex sampling approaches from a correctly-specified selective density in order to conduct inference. Panigrahi et al. [2024], Huang et al. [2024], and Bakshi et al. [2024] extend these ideas to incorporate randomization.

We take our working model for $Y_i \in \mathbb{R}$ to be a GLM with a canonical link function:

$$f_\theta(Y_i) = \exp\left(\frac{Y_i X_i^\top \theta - b(X_i^\top \theta)}{\alpha} + c(Y_i, \alpha)\right), \tag{4.1}$$

where $X_i \in \mathbb{R}^p$ are fixed regressors, $b : \mathbb{R} \mapsto \mathbb{R}$, and α is an over-dispersion parameter for the variance.

Minimizing the negative log-likelihood $-\log f_\theta(Y_i)$ for the working model is equivalent to minimizing the loss

$$\ell_\theta(Y_i) := -Y_i X_i^\top \theta + b(X_i^\top \theta), \tag{4.2}$$

which yields the score variable

$$\hat{Z}_n := \sqrt{n} P_n \dot{\ell}_{\check{\theta}_n^{E,*}} = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} X_i \left(\dot{b}(X_i^\top \check{\theta}_n^{E,*}) - Y_i \right),$$

where $\check{\theta}_n^{E,*}$ was defined in (2.7).

Our goal is valid post-selection inference with a GLM working model and an L_1 penalty for selection. We already established in Proposition 3 that the L_1 penalty satisfies Condition 5. Furthermore, examining the GLM loss function in (4.2), it is clear that Condition 7 is satisfied with $A_i = -X_i$ and $a_{i,\theta} = b\left(X_i^\top \theta\right)$, and so valid inference can be obtained by adding noise to the outcomes as outlined in Corollary 7. It remains to establish that the remaining conditions for Corollary 7 hold.

Since X_i is fixed,

$$\Sigma_n := \mathrm{Var}\left(\hat{Z}_n\right) = \frac{1}{n} \sum_{i=1}^{n} \mathrm{Var}(Y_i) X_i X_i^\top. \tag{4.3}$$

If $X^\top X$ is positive definite, then Σ_n is invertible, satisfying Condition 1(i). Condition 1(ii) holds under mild regularity conditions. Condition 4 involves a Taylor expansion of the loss around $\theta_n^{E,*}$ and holds under standard assumptions for the theory of M-estimation [van der Vaart, 1998, Section 5.2], as established by the following result.

Proposition 8. *Suppose that ℓ is thrice differentiable in θ, and θ is restricted to a compact subset $\Theta \subset \mathbb{R}^p$ over which we have uniform convergence,*

$$\sup_{\theta \in \Theta} \left[P_n \ell_\theta - \mathbb{E} P_n \ell_\theta \right] = o_p(1).$$

Furthermore, suppose that $\theta_n^{E,*}$ in (2.7) is a unique minimizer, that $\mathbb{E} P_n \ell_\theta$ has a unique minimizer θ_n^*, and that we are in the local alternative setting where $\sqrt{n}\theta_n^* = O(1)$. Then under regularity conditions [van der Vaart, 1998, Newey, 1991], Condition 4 holds for $H_n = H_n\left(\check{\theta}_n^{E,*}\right) := \mathbb{E} P_n \ddot{\ell}_{\check{\theta}_n^{E,*}}$.

In the case of a GLM, the Hessian $H_n\left(\check{\theta}_n^{E,*}\right)$ in Proposition 8 takes the form

$$H_n(\theta) := \mathbb{E}P_n\ddot{\ell}_\theta = P_n\ddot{\ell}_\theta = \frac{1}{n}\sum_{i=1}^{n}\ddot{b}(X_i^\top\theta)X_iX_i^\top. \tag{4.4}$$

By construction of the working model (4.1), $\ddot{b}(X_i^\top\theta) > 0$ since under the working model $\mathrm{Var}\,(Y_i) = \alpha\ddot{b}(X_i^\top\theta)$. Therefore, provided that $X^\top X$ has full rank, the Hessian is positive definite. This establishes Condition 2, i.e., the uniqueness conditions of $\bar{\theta}_n^E$ and $\bar{\theta}_n^{(\lambda)}$, along with the uniqueness conditions of Proposition 8 (see Tibshirani [2013] for a more detailed analysis of the settings under which L_1-penalized problems have unique solutions).

Next, we turn to Condition 6, which involves estimating Σ_n defined in (4.3) using an estimator $\bar{\Sigma}_{n,i}$ of $\mathrm{Var}(Y_i)$. One approach, as taken by Huang et al. [2024], is to assume that the the working model (4.1) holds for some particular θ_n^*. Then $\mathrm{Var}\,(Y_i) = \alpha\ddot{b}(X_i^\top\theta_n^*)$ and $E(Y_i) = \dot{b}(X_i^\top\theta_n^*)$. Define the over-dispersion estimator

$$\hat{\alpha}(\theta) = \frac{1}{n-p}\sum_{i=1}^{n}\frac{\left(Y_i - \dot{b}(X_i^\top\theta)\right)^2}{\ddot{b}(X_i^\top\theta)} \tag{4.5}$$

Assuming that the working model holds, classic GLM results provide a consistent estimator $\tilde{\theta}_n$ of θ_n^*, which in turn provides a consistent estimator $\hat{\alpha}\left(\tilde{\theta}_n\right)\ddot{b}\left(X_i^\top\tilde{\theta}_n\right)$ of $\mathrm{Var}(Y_i)$. Under stronger regularity conditions, i.e., uniform integrability or convergence in distribution, consistency can be strengthened to L_1 consistency required for Condition 6 to hold [van der Vaart, 1998, Ch. 2.5]. Alternatively, nonparametric variance estimation approaches exist, e.g. using kernel-based regression methods [Hall and Carroll, 1989, Fan and Yao, 1998].

Finally, we turn to Condition 3, which requires a variance estimator $\hat{S}_n^E$ satisfying

$$\lim_{n\to\infty}\Pr\left(\xi^\top\theta_n^{E,*} \in \mathrm{CI}_n^{\xi,\alpha}\left(\bar{\theta}_n^E, \hat{S}_n^E\right)\right) \geq 1-\alpha. \tag{4.6}$$

Since $\bar{\theta}_n^E$ is the M-estimator in (3.5), where E defines a fixed model, (4.6) is satisfied by the classic model-robust sandwich estimator $\hat{S}_n^E := \left(H_{n,E,E}(\bar{\theta}_n^E)\right)^{-1}\tilde{\Sigma}_{n,E,E}\left(H_{n,E,E}(\bar{\theta}_n^E)\right)^{-1}$ [White, 1980], where $H_{n,E,E}(\theta)$ is a submatrix of the Hessian defined in (4.4) and $\tilde{\Sigma}_{n,E,E}$ is

a diagonal matrix of squared residuals $\left(\left(Y_i - \frac{1}{\gamma}\bar{W}_{n,i}\right) - \dot{b}(X_{i,E}^\top \bar{\theta}_n^E)\right)^2$.

This establishes all of the conditions necessary for application of Corollary 7, whose full procedure is outlined in Algorithm 1. A similar procedure implementing Theorem 6 is given in Section C.1.

Algorithm 1: implementing Corollary 7 for L_1-penalized GLMs

Input: Data $\{Y_i, X_i\}$, penalty $\lambda > 0$, information split $\gamma > 0$.

1. $\tilde{\theta}_n \leftarrow \arg\min_{\theta \in \mathbb{R}^p} P_n \ell_\theta$.
2. Sample noise $\bar{W}_{n,i} \leftarrow \mathrm{N}\left(0, \hat{\alpha}(\tilde{\theta}_n)\ddot{b}\left(X_i^\top \tilde{\theta}_n\right)\right)$.
3. Estimate $\bar{\theta}_n^{(\lambda)}$ as in (3.4) and define $E := \mathrm{supp}\left(\bar{\theta}_n^{(\lambda)}\right)$.
4. Estimate $\bar{\theta}_n^E$ as in (3.5).
5. Compute sandwich variance estimator using outcomes $\left\{Y_i - \frac{1}{\gamma}\bar{W}_{n,i}\right\}_{i=1}^n$:

$$\hat{S}_n^E \leftarrow \left(H_{n,E,E}\left(\bar{\theta}_n^E\right)\right)^{-1} \left[\frac{1}{n}\sum_{i=1}^n X_{i,E}\left(Y_i - \frac{1}{\gamma}\bar{W}_{n,i} - \dot{b}(X_{i,E}^\top \bar{\theta}_n^E)\right)^2 X_{i,E}^\top\right] \left(H_{n,E,E}\left(\bar{\theta}_n^E\right)\right)^{-1}.$$

Output: A confidence interval centered at $\bar{\theta}_n^E$:

$$\mathrm{CI}_n^{\xi,\alpha}\left(\bar{\theta}_n^E, \hat{S}_n^E\right) = \left[\xi^\top \bar{\theta}_n^E \pm n^{-1/2} z_{1-\alpha/2}\left(\xi^\top \hat{S}_n^E \xi\right)^{1/2}\right].$$

5 Experimental results

We now empirically study the procedure outlined in Algorithm 1, first in simulation and then on the *Teenage Friends and Lifestyle Study* dataset. All code is provided at https://github.com/rflperry/m_estimation_SI.

5.1 Simulation studies: comparisons to data thinning

In the following simulations, we generated outcomes from a generalized linear model with canonical link function, i.e., $\mathbb{E}[Y_i] = g^{-1}\left(X_i^\top \theta_n^*\right)$. The covariates $X_i \in \mathbb{R}^{40}$ are normally distributed with variance one and an equicorrelation of 0.3. The true mean function θ_n^* is a sparse vector with ten nonzero entries. Selection is performed using the lasso penalty $\rho_\lambda(\theta) := \lambda \|\theta\|_1$, where $\lambda \propto \sqrt{(1+\gamma^2)\log(p)}\,\mathrm{sd}(Y_1, \ldots, Y_n)$; recall from (3.4) that this penalty is then further scaled by $n^{-1/2}$. We set $\gamma = 1$, which effectively allocates half of the information in the data for selection and half for inference.

Our first simulation investigates the setting of a linear mean and homoskedastic Gaussian noise with an unknown variance, i.e., $Y_i = X_i^\top \theta_n^* + \varepsilon_i$. We apply Algorithm 1 to conduct inference on the model selected using L_1-penalized ordinary least squares. In this setting, score thinning closely aligns with the Gaussian data thinning proposals of Rasines and Young [2023] (with a different variance estimator) and Lei [2025]. We compare score thinning to the randomized conditional selective inference approaches of Panigrahi et al. [2024] and Huang et al. [2024], both of which have the *exact* same selection procedure as score thinning but rely on bespoke conditional likelihoods for inference. Lastly, we include comparisons to sample splitting (which uses half of the samples for selection and half for inference, in order to match the information split of $\gamma = 1$) and the classical approach (which reuses the data for selection and inference).

In the left-hand panel of Figure 1, we display the average coverage of 90% confidence intervals for all parameters in each selected model; a method that controls the conditional coverage in the sense of (2.6) will attain 90% coverage. The classical method (which reuses the data for selection and inference) fails to achieve the nominal coverage, and thus we exclude it from the remaining panels. All other approaches attain 90% coverage, except that of Huang et al. [2024], which approximates a selective MLE and has been shown to undercover in practice [Panigrahi et al., 2024]. In contrast, Panigrahi et al. [2024] employ an *exact* approach that attains the nominal coverage. In the middle panel of Figure 1, we assess the power

of each method by comparing the distribution of average confidence interval widths. Score thinning yields wider intervals on average than the randomized conditional inference approaches of Panigrahi et al. [2024] and Huang et al. [2024], which is expected since the conditional approaches are able to use more information for inference and since Huang et al. [2024] does not attain the nominal coverage. Score thinning yields narrower intervals than sample splitting, as suggested by Proposition 1 of Rasines and Young [2023]. The poor performance of sample splitting may also be due to the fact that sample splitting can lead to a poorly-conditioned design matrix when n is small. Lastly, we assess the model selection accuracy in the right-hand panel of Figure 1 by comparing the average false discovery rate (FDR) of each selective inference method. The FDR of a model E is given by $|E \setminus \text{supp}\,(\theta_n^*)|\,/|E|$, i.e., the fraction of incorrectly selected features relative to the total number of selected features. As mentioned earlier, score thinning and the randomized conditional approaches of Huang et al. [2024] and Panigrahi et al. [2024] have identical selection events and therefore have identical FDRs, which are comparable to that of sample splitting; all methods select models of a similar size $|E|$.

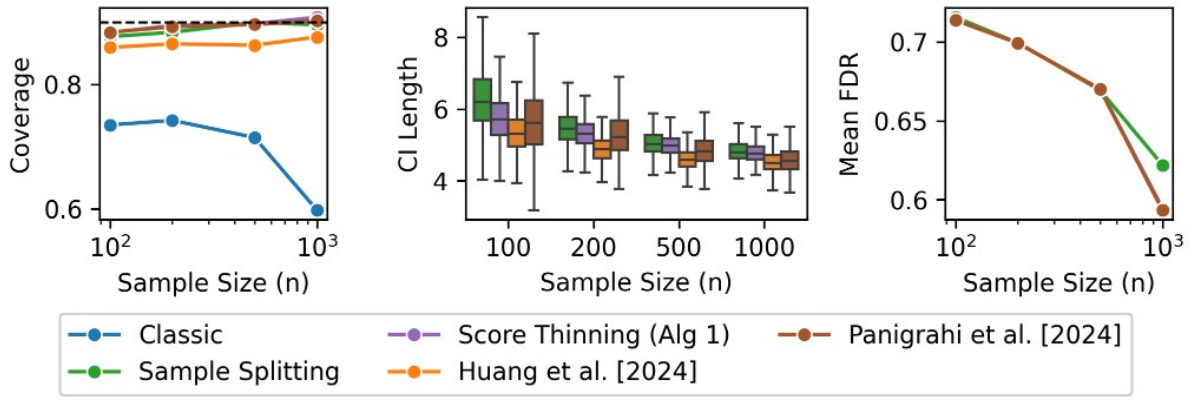


Figure 1: Outcomes are generated from a linear model with i.i.d. Gaussian noise whose variance is unknown. Inference is conducted on the linear model selected by L_1-penalized ordinary least squares. Results are aggregated across 1000 simulated datasets. Since the classical approach (which reuses the data for selection and inference) fails to attain valid coverage, we omit it from the center and right-hand panels.

In our second simulation, outcomes are generated from a Poisson regression model, with and without Gamma-distributed overdisperion. With overdispersion, the outcomes are marginally negative binomial (NB). Inference is conducted on

the model selected using L_1-penalized Poisson regression. Since the outcomes are Poisson or NB, data thinning can be applied to obtain datasets for selection and inference that are *exactly* independent [Neufeld et al., 2024]. Score thinning also applies. In the top row of Figure 2 (in the absence of overdispersion), we see that all of the selective inference methods attain nominal coverage and perform similarly. However, in the bottom row (in the presence of overdispersion), we see that Poisson thinning fails to attain nominal coverage; this is because applying Poisson thinning to NB data does not yield independent datasets. NB thinning and our score thinning proposal have comparable performances; however, while the former requires correct specification of the data-generating distribution, the latter requires much weaker assumptions.

5.2 Simulation studies: comparisons beyond data thinning

We now move to settings in which data thinning *cannot be applied* to obtain *exactly* independent datasets for selection and inference. We first consider outcomes generated from a logistic regression model. Inference is conducted on the model selected using L_1-penalized logistic regression. Since the outcomes are Bernoulli, it is not possible to thin them into independent datasets [Dharamshi et al., 2025], although score thinning still applies. The left-hand panel of Figure 3 shows that the classical method (which reuses the data for selection and inference) fails to attain valid coverage. The coverage of Huang et al. [2024] once again falls below the nominal level. The exact approach of Panigrahi et al. [2024] is not displayed as it is only applicable to the linear Gaussian setting. In principle, the exact approach of Bakshi et al. [2024] could be applied, although they do not provide software for their extension to logistic regression. In the center panel of Figure 3, the randomized conditional method of Huang et al. [2024] has the narrowest intervals, possibly because they fail to attain the nominal coverage. Sample splitting yields the widest intervals, again likely due to the ill-conditioning of the design matrix. In the right-hand panel of Figure 3, the FDRs are comparable.

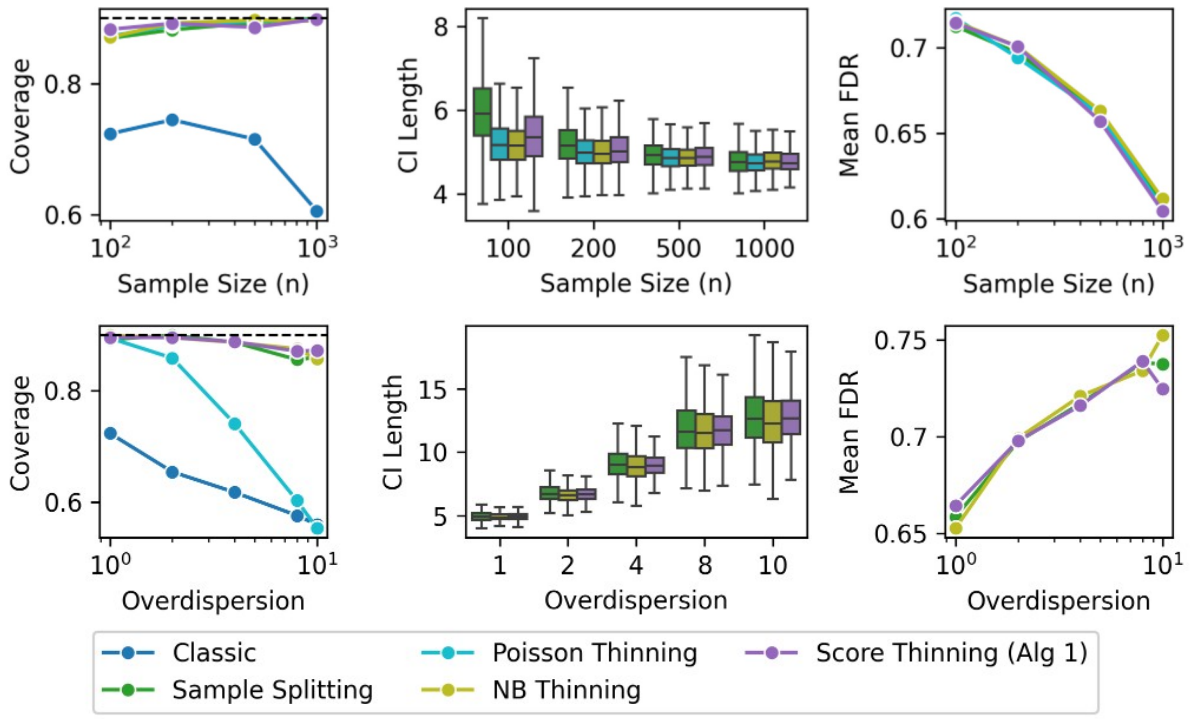


Figure 2: We simulate outcomes from a Poisson regression model with and without overdispersion. Inference is conducted on the model selected using L_1-penalized Poisson regression. Confidence intervals are computed for all coefficients in the selected model. Results are aggregated across 1000 simulated datasets. Since the classical approach (which reuses the data for selection and inference) fails to attain nominal coverage, we omit it from the center and right-hand panels. **(Top):** In the absence of overdispersion, as we vary the sample size, all thinning methods perform comparably. **(Bottom):** We vary the overdispersion of the variance relative to the mean; this induces a negative binomial outcome distribution. Since Poisson thinning fails to attain valid coverage, we omit it from the center and right-hand panels.

We note that existing software for (penalized) logistic regression, e.g, `glmnet` in `R` [Friedman et al., 2010], typically requires that the inputted outcome values are binary or at least non-negative. Because the noisy outcomes in (3.4) and (3.5) are non-binary and contain negative values, we cannot use `glmnet` out of the box to solve (3.4) or (3.5). Instead, our software implementation makes use of the `Python` package `RegReg` [Klingenberg and Taylor, 2021].

In our final simulation, we consider a more involved setting for which no existing methods are available in the selective inference literature: cluster-correlated data. We simulate observations whose mean is linear in the covariates $X_{i,j}$ but which have a cluster-specific intercept, i.e., $Y_{i,j} = X_{i,j}^\top \theta_n^* + \nu_i + \varepsilon_{i,j}$ where $\nu_i \sim \mathrm{N}(0, 0.5)$, $\varepsilon_{i,j}$

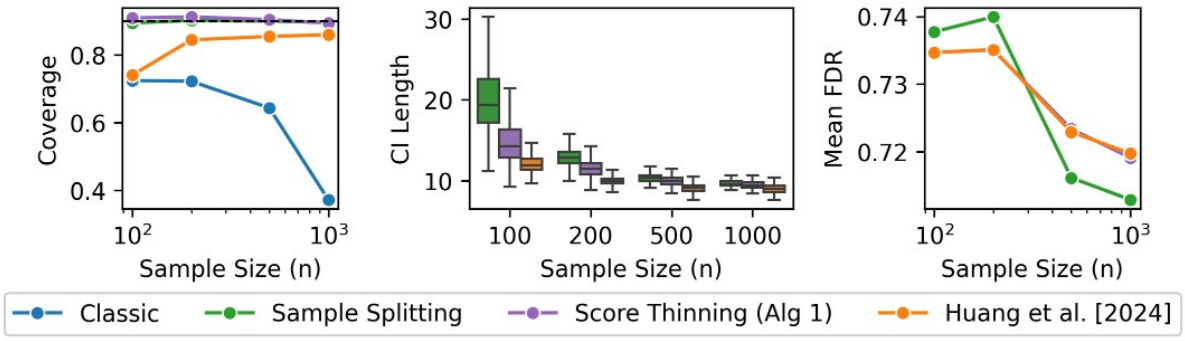


Figure 3: We simulate outcomes from a logistic regression model. Inference is conducted on the model selected using L_1-penalized logistic regression. Confidence intervals are computed for all coefficients in the selected model. Results are aggregated across 1000 simulated datasets. Since the classical approach (which reuses the data for selection and inference) fails to attain valid coverage, we omit it from the center and right-hand panels.

are i.i.d. symmetric Laplace random variables with variance 0.5, $j = 1, \ldots, 10$, and $i = 1, \ldots, n/10$; at present, no method is available to thin a Laplace random variable into independent components. Thus, n observations are grouped into equicorrelated clusters of size ten. Algorithm 1 is not immediately applicable to this setting since the working model in (4.1) assumed independent outcomes rather than cluster-correlated outcomes. We therefore apply Algorithm 1 with minor modifications: in Step 2 we estimate the marginal variance and equicorrelation parameter, and in Step 5 we use a sandwich variance estimator with an equicorrelated working model. Figure 4 shows that the classical approach (which reuses the data for selection and inference) again fails to achieve the nominal coverage, while score thinning attains narrower confidence intervals than sample splitting. We note that sample splitting is performed on the cluster level in order for the independence of the selection and inference steps to be preserved. We do not compare to a randomized conditional approach as no existing proposal can accommodate this setting.

5.3 Application to principal components network regression

The *Teenage Friends and Lifestyle Study* dataset contains the survey responses of secondary school students in Glasgow [Michell and West, 1996]. The students were

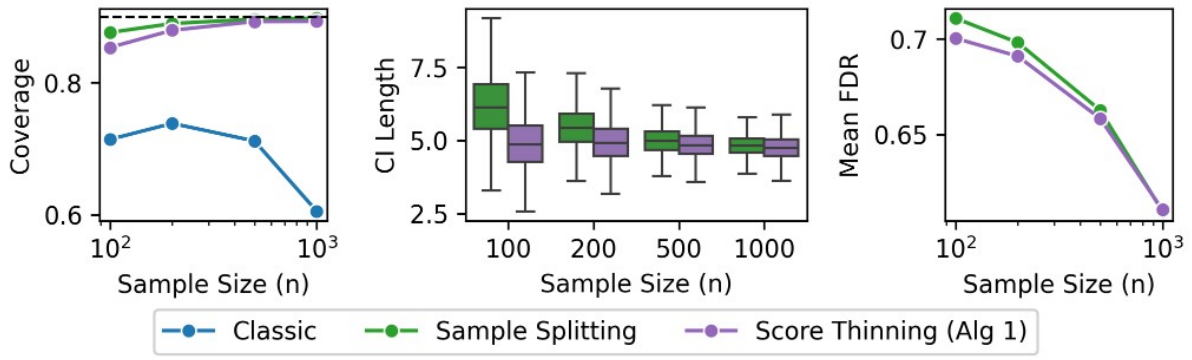


Figure 4: Outcomes are generated from a linear model with equicorrelated non-Gaussian noise. Inference is conducted in the linear model selected by L_1-penalized linear regression. Results are aggregated across 5000 simulated datasets. Since the classical approach (which reuses the data for selection and inference) fails to attain valid coverage, we omit it from the center and right-hand panels.

surveyed three times between 1995 and 1997, and asked various questions about their lifestyle including their smoking usage and who their friends were. Thus, the observations lie on an observed friendship network.

Using the first wave of survey responses, Hayes et al. [2025] estimate the association between the sex of the student and their odds of smoking. Without accounting for the network, a logistic regression model accounting for age and church attendance estimates the odds ratio of being a smoker when male rather than female to be 0.30 with a 95% confidence interval of $(0.11, 0.81)$. However, we can see from the left and middle panels of Figure 5 that there are groups of friends who smoke.

To determine whether the association between sex and smoking persists after accounting for the friendship network, Hayes et al. [2025] further include the singular vectors from the singular value decomposition of the network's adjacency matrix as regression covariates. However, in doing so they face two uncertain modeling choices. First, since the graph is directed, the adjacency matrix is non-symmetric and thus the left and right singular vectors are different. Specifically, the former relate to outgoing edges in the network while the latter relate to incoming edges. Second, the number of singular vectors r to include in the regression is unknown, and implies an assumption on the rank of the network. The authors arbitrarily choose to include only the right singular vectors in the regression, and then present

results for $r = 1, \ldots, 25$.

To adaptively choose a model, we employ score thinning in Algorithm 1 (using an L_1-penalized logistic regression model with $\gamma = 1$ and the same λ as in the simulations). The $n \times p$ design matrix X consists of $n = 152$ subjects and $p = 56$ features, which consist of an intercept, age, sex, church attendance (four categories), and all of the first 25 left and right singular vectors. Since the singular vectors used to construct X are based on the entire network, sample splitting is not applicable. The noisy outcomes used for selection and inference are plotted in the right-hand panel of Figure 5.

The singular vectors selected via L_1-penalized logistic regression using the noisy responses include a mix of non-consecutive left and right singular vectors. The estimated odds ratio of being a smoker when male rather than female is 0.43 with a 95% confidence interval of $(0.06, 3.1)$. Hayes et al. [2025] similarly found non-significant estimates, in line with the hypothesis that the apparent association between sex and smoking is the spurious result of the network structure.

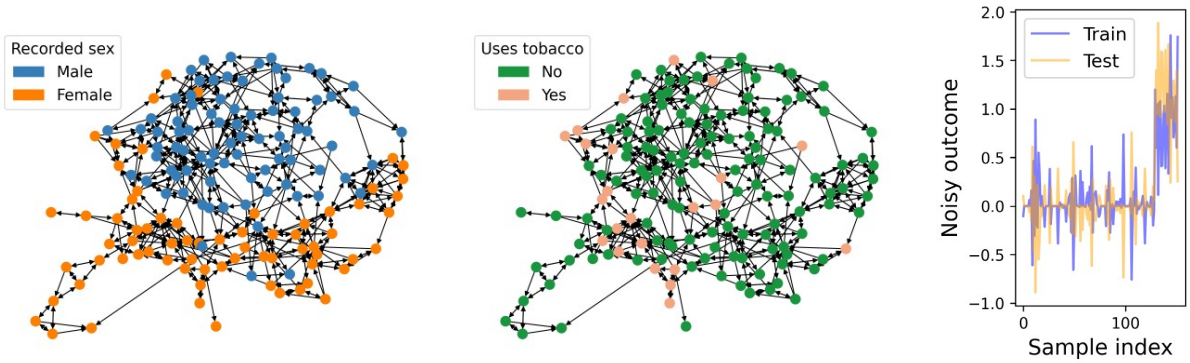


Figure 5: (Left): the friendship network of students, colored by recorded sex. (Middle): the friendship network of students, colored by their reported tobacco usage Y. (Right): the noisy outcomes constructed by adding and subtracting Gaussian noise to the outcomes Y, i.e., $Y_i + \bar{W}_{n,i}$ and $Y_i - \bar{W}_{n,i}$. These sets are not independent, but selection and inference using them are asymptotically independent.

6 Discussion

Existing approaches to selective inference trade off statistical efficiency, ease of implementation, flexibility of the selection procedure, and distributional assump-

tions. Fully conditional approaches maximize the statistical efficiency in the sense of Fithian et al. [2017] by conditioning on the minimal amount of information; however, these methods require bespoke derivations and computational implementations for each selection procedure [Lee et al., 2016, Tian and Taylor, 2018, Panigrahi et al., 2024, Huang et al., 2024]. On the other hand, applying data thinning to the response to obtain exactly independent datasets for selection and inference can accommodate arbitrary selection procedures and is easy to implement; yet, data thinning is less efficient and requires strong distributional assumptions [Rasines and Young, 2023, Neufeld et al., 2024, Dharamshi et al., 2025].

We have introduced score thinning, which is situated in between these two classes of approaches. Our results suggest that adding and subtracting Gaussian noise — that is, conducting "Gaussian data thinning" on non-Gaussian data — may be somewhat robust to model misspecification in penalized M-estimation post-selection inference tasks, provided that the sample size is sufficiently large.

Furthermore, our work demonstrates an equivalence between adding and subtracting noise from the outcome, as in Algorithm 1, and adding and subtracting noise from the gradient, as in Algorithm 2 and other selective inference papers [Tian and Taylor, 2018, Panigrahi et al., 2024, Huang et al., 2024]. While the latter may be more widely applicable to penalized M-estimation problems, the former is often more practical as it avoids the need for bespoke software.

Our work suggests many new questions. First, our theoretical results are limited to the setting where p is fixed, and yet the setting where p is growing is also of substantial interest. Second, while Condition 5 is satisfied by the L_1 penalty, it remains to show that it holds for other popular penalties such as the group lasso. Lastly, the idea of thinning functions of the data, rather than the data itself, is a widely applicable idea which may allow for further development of strategies inspired by data thinning in the absence of distributional assumptions.

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References

Soham Bakshi, Walter Dempsey, and Snigdha Panigrahi. Selective inference for time-varying effect moderation. *arXiv*, November 2024. arXiv:2411.15908.

Yoav Benjamini, Ruth Heller, and Daniel Yekutieli. Selective inference in complex research. *Philosophical Transactions. Series A, Mathematical, Physical, and Engineering Sciences*, 367(1906):4255–4271, November 2009.

Richard Berk, Lawrence Brown, Andreas Buja, Kai Zhang, and Linda Zhao. Valid post-selection inference. *The Annals of Statistics*, 41(2):802–837, April 2013.

Louis H. Y. Chen and Xiao Fang. Multivariate Normal Approximation by Stein's Method: The Concentration Inequality Approach. *arXiv*, May 2015. arXiv:1111.4073.

D. R. Cox. A note on data-splitting for the evaluation of significance levels. *Biometrika*, 62(2):441–444, August 1975.

Luc Devroye, Abbas Mehrabian, and Tommy Reddad. The total variation distance between high-dimensional Gaussians with the same mean. *arXiv*, October 2023. arXiv:1810.08693.

Ameer Dharamshi, Anna Neufeld, Keshav Motwani, Lucy L. Gao, Daniela Witten, and Jacob Bien. Generalized Data Thinning Using Sufficient Statistics. *Journal of the American Statistical Association*, 120(549):511–523, January 2025.

Jianqing Fan and Qiwei Yao. Efficient Estimation of Conditional Variance Functions in Stochastic Regression. *Biometrika*, 85(3):645–660, 1998.

William Fithian, Dennis Sun, and Jonathan Taylor. Optimal Inference After Model Selection. *arXiv*, April 2017. arXiv:1410.2597.

Jerome Friedman, Trevor Hastie, and Robert Tibshirani. Regularization paths for generalized linear models via coordinate descent. *Journal of Statistical Software*, 33(1):1–22, 2010.

Kevin Fry, Panigrahi , Snigdha, , and Jonathan Taylor. Assumption-Lean Data Fission with Resampled Data. *Journal of the American Statistical Association*, 120(549):161–161, January 2025.

Peter Hall and R. J. Carroll. Variance Function Estimation in Regression: The Effect of Estimating the Mean. *Journal of the Royal Statistical Society. Series B (Methodological)*, 51(1):3–14, 1989.

Alex Hayes, Mark M. Fredrickson, and Keith Levin. Estimating Network-Mediated Causal Effects via Principal Components Network Regression. *Journal of Machine Learning Research*, 26(13):1–99, 2025.

Harold Hotelling. The selection of variates for use in prediction with some comments on the general problem of nuisance parameters. *The Annals of Mathematical Statistics*, 11(3):271–283, 1940.

Yiling Huang, Sarah Pirenne, Snigdha Panigrahi, and Gerda Claeskens. Selective inference using randomized group lasso estimators for general models. *arXiv*, March 2024. arXiv:2306.13829.

Adel Javanmard and Andrea Montanari. Confidence Intervals and Hypothesis Testing for High-Dimensional Regression. *Journal of Machine Learning Research*, 15 (82):2869–2909, 2014.

Danijel Kivaranovic and Hannes Leeb. On the Length of Post-Model-Selection Confidence Intervals Conditional on Polyhedral Constraints. *Journal of the American Statistical Association*, 116(534):845–857, April 2021.

B. Klingenberg and J. Taylor. Regreg, 2021. https://github.com/regreg/regreg.

Nikolaus Kriegeskorte, W Kyle Simmons, Patrick SF Bellgowan, and Chris I Baker. Circular analysis in systems neuroscience – the dangers of double dipping. *Nature Neuroscience*, 12(5):535–540, May 2009.

Arun K Kuchibhotla, John E Kolassa, and Todd A Kuffner. Post-selection inference. *Annual Review of Statistics and Its Application*, 9(1):505–527, 2022.

Jason D. Lee, Dennis L. Sun, Yuekai Sun, and Jonathan E. Taylor. Exact post-selection inference, with application to the lasso. *The Annals of Statistics*, 44(3): 907 – 927, June 2016.

Lihua Lei. Discussion of "Data Fission: Splitting a Single Data Point" – Some Asymptotic Results for Data Fission. *Journal of the American Statistical Association*, 120(549):147–150, January 2025.

James Leiner, Boyan Duan, Larry Wasserman, and Aaditya Ramdas. Data Fission: Splitting a Single Data Point. *Journal of the American Statistical Association*, 120(549):135–146, January 2025.

Nicolai Meinshausen, Lukas Meier, and Peter Bühlmann. p-Values for High-Dimensional Regression. *Journal of the American Statistical Association*, 104 (488):1671–1681, December 2009.

L. Michell and P. West. Peer pressure to smoke: the meaning depends on the method. *Health Education Research*, 11(1):39–49, March 1996.

Fedor Nazarov. On the maximal perimeter of a convex set in $\mathbb{R}^n$ with respect to a gaussian measure. In *Geometric Aspects of Functional Analysis*, pages 169–187. Springer, January 2003.

Anna Neufeld, Ameer Dharamshi, Lucy L. Gao, and Daniela Witten. Data Thinning for Convolution-Closed Distributions. *Journal of Machine Learning Research*, 25 (57):1–35, 2024.

Anna Neufeld, Ameer Dharamshi, Lucy L. Gao, Daniela Witten, and Jacob Bien. Discussion of "Data fission: splitting a single data point". *Journal of the American Statistical Association*, 120(549):151–157, 2025.

Anna Neufeld, Ronan Perry, and Daniela Witten. Inference conditional on selection: a review, April 2026. arXiv:2604.09779 [stat.ME].

Whitney K. Newey. Uniform Convergence in Probability and Stochastic Equicontinuity. *Econometrica*, 59(4):1161–1167, 1991.

Natalia L. Oliveira, Jing Lei, and Ryan J. Tibshirani. Unbiased risk estimation in the normal means problem via coupled bootstrap techniques. *Electronic Journal of Statistics*, 18(2):5405–5448, January 2024.

Snigdha Panigrahi, Kevin Fry, and Jonathan Taylor. Exact selective inference with randomization. *Biometrika*, 111(4):1109–1127, December 2024.

Martin Raič. A multivariate Berry–Esseen theorem with explicit constants. *Bernoulli*, 25(4A), November 2019.

D García Rasines and G A Young. Splitting strategies for post-selection inference. *Biometrika*, 110(3):597–614, September 2023.

James Robins and Aad van der Vaart. Adaptive nonparametric confidence sets. *The Annals of Statistics*, 34(1):229–253, February 2006.

Souhardya Sengupta and Lucas Janson. The ℓ-test: leveraging sparsity in the Gaussian linear model for improved inference, November 2025. arXiv:2406.18390 [stat].

Xiaoying Tian. Prediction error after model search. *The Annals of Statistics*, 48 (2):763–784, April 2020.

Xiaoying Tian and Jonathan Taylor. Asymptotics of Selective Inference. *Scandinavian Journal of Statistics*, 44(2):480–499, 2017.

Xiaoying Tian and Jonathan Taylor. Selective Inference with a Randomized Response. *The Annals of Statistics*, 46(2):679–710, 2018.

Robert Tibshirani. Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 58(1):267–288, 1996.

Ryan J. Tibshirani. The lasso problem and uniqueness. *Electronic Journal of Statistics*, 7:1456–1490, January 2013.

Ryan J. Tibshirani, Alessandro Rinaldo, Rob Tibshirani, and Larry Wasserman. Uniform asymptotic inference and the bootstrap after model selection. *The Annals of Statistics*, 46(3):1255–1287, June 2018.

Aad van der Vaart. *Asymptotic statistics*. Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge University Press, 1998.

Halbert White. A Heteroskedasticity-Consistent Covariance Matrix Estimator and a Direct Test for Heteroskedasticity. *Econometrica*, 48(4):817–838, 1980.

James Yang and Trevor Hastie. A Fast and Scalable Pathwise-Solver for Group Lasso and Elastic Net Penalized Regression via Block-Coordinate Descent, May 2024. arXiv:2405.08631 [stat.CO].

Cun-Hui Zhang and Stephanie S. Zhang. Confidence Intervals for Low Dimensional Parameters in High Dimensional Linear Models. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 76(1):217–242, January 2014.

A Supporting theory and their proofs

We state and prove technical results necessary for proving results in the main text.

A.1 An extended Berry-Esseen bound

We first recall the multivariate Berry-Esseen Theorem.

Theorem A.1 (Berry-Esseen [Raič, 2019])**.** *Let U_i be independent mean zero random variables in $\mathbb{R}^p$ and let $V_n := \sum_{i=1}^n U_i$ such that $\mathrm{Var}(V_n) = \sum_{i=1}^n \mathrm{Var}(U_i) = I_p$. Define $V \sim \mathrm{N}(0, I_p)$. For any measurable and convex set $A \subseteq \mathbb{R}^p$,*

$$|\Pr(V_n \in A) - \Pr(V \in A)| \leq \left(42p^{1/4} + 16\right) \sum_{i=1}^n \mathbb{E}\,\|U_i\|_2^3 .$$

Equivalently, this result says that

$$|\Pr(V_n \in A) - \Pr(V \in A)| = O\left(p^{1/4} \sum_{i=1}^n \mathbb{E}\,\|U_i\|_2^3\right)$$

for any set $A \subseteq \mathbb{R}^p$ that is convex and measurable.

The following lemma follows from Theorem A.1.

Lemma A.2. *Let U_i be independent random variables in $\mathbb{R}^p$ and assume that $\mu_{n,i} := \mathbb{E}[U_i]$ is finite. Define $\hat{V}_n := n^{-1/2} \sum_{i=1}^n U_i$ and assume $\Sigma_n := \mathrm{Var}(\hat{V}_n) = \frac{1}{n} \sum_{i=1}^n \mathrm{Var}(U_i)$ is invertible. Define $\mu_n := \frac{1}{n} \sum_{i=1}^n \mu_{n,i}$ and let $V_n \sim \mathrm{N}(\sqrt{n}\mu_n, \Sigma_n)$. Then for any set $A \subseteq \mathbb{R}^p$ which can be expressed as a union of K measurable almost surely disjoint and convex sets, it holds that*

$$\left| \Pr\left(\hat{V}_n \in A\right) - \Pr\left(V_n \in A\right) \right| = O\left(K p^{1/4} n^{-3/2} \lambda_{\min}^{-3/2}(\Sigma_n) \sum_{i=1}^n \mathbb{E} \left\| U_i - \mu_{n,i} \right\|_2^3 \right).$$

Proof.

Define $\tilde{A} := \left\{ \Sigma_n^{-1/2}(a - \sqrt{n}\mu_n) : a \in A \right\}$ (omitting a dependence on n for brevity), and note that

$$\begin{aligned} &\left| \Pr\left(\hat{V}_n \in A\right) - \Pr\left(V_n \in A\right) \right| \\ &= \left| \Pr\left(\Sigma_n^{-1/2} \left(\hat{V}_n - \sqrt{n}\mu_n \right) \in \tilde{A} \right) - \Pr\left(\Sigma_n^{-1/2} \left(V_n - \sqrt{n}\mu_n \right) \in \tilde{A} \right) \right|. \end{aligned} \tag{A.1}$$

The term

$$\Sigma_n^{-1/2} \left(\hat{V}_n - \sqrt{n}\mu_n \right) = \sum_{i=1}^n n^{-1/2} \Sigma_n^{-1/2} \left(U_i - \mu_{n,i} \right)$$

is the sum of mean-zero and independent observations $n^{-1/2}\Sigma_n^{-1/2}(U_i - \mu_{n,i})$. Furthermore, $\mathrm{Var}\left(\Sigma_n^{-1/2} \left(\hat{V}_n - \sqrt{n}\mu_n \right) \right) = I_p$ and $\Sigma_n^{-1/2} \left(V_n - \sqrt{n}\mu_n \right) \sim \mathrm{N}(0, I_p)$.

By assumption, $\tilde{A}$ remains a union of at most K almost surely disjoint, measurable, and convex sets, since the transformation induced by $\Sigma_n^{-1/2}$ and μ_n preserves these properties. So, we can write $\tilde{A} = \bigcup_{k=1}^K \tilde{A}_k$ where $\left\{ \tilde{A}_k \right\}_{k=1}^K$ are almost surely

disjoint, measurable, and convex. It follows that

$$\begin{aligned}
&\left|\Pr\left(\Sigma_n^{-1/2}\left(\hat{V}_n - \sqrt{n}\mu_n\right) \in \tilde{A}\right) - \Pr\left(\Sigma_n^{-1/2}\left(V_n - \sqrt{n}\mu_n\right) \in \tilde{A}\right)\right| \\
&= \left|\sum_{k=1}^{K}\left[\Pr\left(\Sigma_n^{-1/2}\left(\hat{V}_n - \sqrt{n}\mu_n\right) \in \tilde{A}_k\right) - \Pr\left(\Sigma_n^{-1/2}\left(V_n - \sqrt{n}\mu_n\right) \in \tilde{A}_k\right)\right]\right| \\
&\leq \sum_{k=1}^{K}\left|\Pr\left(\Sigma_n^{-1/2}\left(\hat{V}_n - \sqrt{n}\mu_n\right) \in \tilde{A}_k\right) - \Pr\left(\Sigma_n^{-1/2}\left(V_n - \sqrt{n}\mu_n\right) \in \tilde{A}_k\right)\right| \\
&\leq \sum_{k=1}^{K} O\left(p^{1/4}\sum_{i=1}^{n}\mathbb{E}\left\|n^{-1/2}\Sigma_n^{-1/2}\left(U_i - \mu_{n,i}\right)\right\|_2^3\right) \\
&= O\left(Kp^{1/4}n^{-3/2}\lambda_{\min}^{-3/2}\left(\Sigma_n\right)\sum_{i=1}^{n}\mathbb{E}\left\|U_i - \mu_{n,i}\right\|_2^3\right).
\end{aligned}$$

Combining this with (A.1), we obtain our bound. □

A.2 Helper lemma for Lemma 1

We begin with a helper lemma that will be used in the proof of Lemma 1.

Lemma A.3. *Suppose that* $W_n \sim \mathrm{N}\left(0, \Sigma_n\right)$ *for* Σ_n *defined in Condition 1, and define*

$$\tilde{Z}_n^{(1)} := \hat{Z}_n + \gamma W_n \quad \text{and} \quad \tilde{Z}_n^{(2)} := \hat{Z}_n - \frac{1}{\gamma}W_n.$$

Under Condition 1, for any set C *which is the union of at most* K *measurable convex sets,*

$$\left|\Pr\left(\left(\tilde{Z}_n^{(1)}, \tilde{Z}_n^{(2)}\right) \in C\right) - \Pr\left(\left(Z_n^{(1)}, Z_n^{(2)}\right) \in C\right)\right| = O\left(2^K p^{1/4} n^{-1/2}\lambda_{\min}^{-3/2}\left(\Sigma_n\right)\right). \tag{A.2}$$

Proof. We wish to apply Lemma A.2, and thus require that $\left(\tilde{Z}_n^{(1)}, \tilde{Z}_n^{(2)}\right)$ are equal in distribution to the sum of independent random variables with a specified covariance

matrix. Let $W_{n,i} \overset{iid}{\sim} \mathrm{N}(0, \Sigma_n)$. First, note that

$$\begin{aligned}\tilde{Z}_n^{(1,2)} := \begin{pmatrix} \tilde{Z}_n^{(1)} \\ \tilde{Z}_n^{(2)} \end{pmatrix} &= n^{-1/2} \sum_{i=1}^{n} \begin{pmatrix} \dot{\ell}_{\tilde{\theta}_n^{E,*}}(X_i, Y_i) \\ \dot{\ell}_{\tilde{\theta}_n^{E,*}}(X_i, Y_i) \end{pmatrix} + \begin{pmatrix} \gamma W_n \\ -\gamma^{-1} W_n \end{pmatrix} \\ &\overset{d}{=} n^{-1/2} \sum_{i=1}^{n} \begin{pmatrix} \dot{\ell}_{\tilde{\theta}_n^{E,*}}(X_i, Y_i) + \gamma W_{n,i} \\ \dot{\ell}_{\tilde{\theta}_n^{E,*}}(X_i, Y_i) - \gamma^{-1} W_{n,i} \end{pmatrix}\end{aligned}$$

is clearly the sum of independent random variables; $\tilde{Z}_n^{(1,2)}$ has mean $\left(\sqrt{n}\mu_n^{E,*}, \sqrt{n}\mu_n^{E,*}\right)$. Second, $\tilde{Z}_n^{(1,2)}$ has covariance matrix

$$\begin{aligned}\mathrm{Var}\left(\begin{pmatrix} \tilde{Z}_n^{(1)} \\ \tilde{Z}_n^{(2)} \end{pmatrix}\right) &= \mathrm{Var}\left(\begin{pmatrix} \tilde{Z}_n \\ \tilde{Z}_n \end{pmatrix} + \begin{pmatrix} \gamma W_n \\ -\gamma^{-1} W_n \end{pmatrix}\right) \\ &= \begin{bmatrix} \Sigma_n & \Sigma_n \\ \Sigma_n & \Sigma_n \end{bmatrix} + \begin{bmatrix} \gamma^2 \Sigma_n & -\Sigma_n \\ -\Sigma_n & \gamma^{-2} \Sigma_n \end{bmatrix} \\ &= \begin{bmatrix} (1+\gamma^2)\Sigma_n & 0 \\ 0 & \left(1+\gamma^{-2}\right)\Sigma_n \end{bmatrix}. \qquad \text{(A.3)}\end{aligned}$$

The covariance matrix exists and is invertible by Condition 1(i), with minimum eigenvalue $\gamma_{\min}\lambda_{\min}(\Sigma_n)$, where $\gamma_{\min} := \min\left\{1+\gamma^2, 1+\gamma^{-2}\right\}$. Recalling the definitions of $Z_n^{(1)}$ and $Z_n^{(2)}$ in (2.11), we note that similarly $Z_n^{(1,2)} := \begin{pmatrix} Z_n^{(1)} \\ Z_n^{(2)} \end{pmatrix}$ is multivariate normal with the same mean and covariance.

Applying Lemma A.2, under Condition 1(ii) which ensures that the centered third moment is uniformly bounded,

$$\left|\Pr\left(\tilde{Z}_n^{(1,2)} \in C\right) - \Pr\left(Z_n^{(1,2)} \in C\right)\right| = O\left(Kp^{1/4} n^{-1/2} \gamma_{\min}^{-3/2} \lambda_{\min}^{-3/2}(\Sigma_n)\right).$$

Since $\gamma_{\min} \geq 1$, it holds that $\gamma_{\min}^{-3/2} \leq 1$ and so we obtain our final bound

$$\left|\Pr\left(\tilde{Z}_n^{(1,2)} \in C\right) - \Pr\left(Z_n^{(1,2)} \in C\right)\right| = O\left(Kp^{1/4} n^{-1/2} \lambda_{\min}^{-3/2}(\Sigma_n)\right).$$

□

B Proofs of main text results

B.1 Proof of Lemma 1

Proof. Define

$$\hat{Z}_n^{(1,2)} := \begin{pmatrix} \hat{Z}_n^{(1)} \\ \hat{Z}_n^{(2)} \end{pmatrix}.$$

Lemma A.3 established a Berry-Esseen-type bound for thinned variables $\tilde{Z}_n^{(1,2)}$ constructed using the *known* variance Σ_n. We use this helper lemma to establish a bound on the thinned variables $\hat{Z}_n^{(1,2)}$ constructed using an *estimate* $\hat{\Sigma}_n$ of Σ_n.

First, by the triangle inequality, and for $\tilde{Z}_n^{(1,2)}$ defined in Lemma A.3,

$$\begin{aligned}
&\left|\Pr\left(\hat{Z}_n^{(1,2)} \in C\right) - \Pr\left(Z_n^{(1,2)} \in C\right)\right| \\
&\leq \left|\Pr\left(\hat{Z}_n^{(1,2)} \in C\right) - \Pr\left(\tilde{Z}_n^{(1,2)} \in C\right)\right| + \left|\Pr\left(\tilde{Z}_n^{(1,2)} \in C\right) - \Pr\left(Z_n^{(1,2)} \in C\right)\right| \\
&= \left|\Pr\left(\hat{Z}_n^{(1,2)} \in C\right) - \Pr\left(\tilde{Z}_n^{(1,2)} \in C\right)\right| + O\left(Kp^{1/4}n^{-1/2}\lambda_{\min}^{-3/2}\left(\Sigma_n\right)\right), \qquad \text{(B.1)}
\end{aligned}$$

where the last line follows from (A.2).

For any $\Delta > 0$, we apply the triangle inequality to the remaining term using the intermediate variable

$$\hat{Z}_n^{(1,2),\Delta} := \begin{cases} \hat{Z}_n^{(1,2)}, & \text{if} \quad \mathbb{I}\left[\left\|\hat{\Sigma}_n\Sigma_n^{-1} - I_p\right\|_F \leq \Delta\right] \\ \tilde{Z}_n^{(1,2)}, & \text{if} \quad \mathbb{I}\left[\left\|\hat{\Sigma}_n\Sigma_n^{-1} - I_p\right\|_F > \Delta\right] \end{cases}.$$

When the estimated variance $\hat{\Sigma}_n$ is "close" to Σ_n, then $\hat{Z}_n^{(1,2),\Delta}$ is equal to $\hat{Z}_n^{(1,2)}$. Otherwise, it is defined to be $\tilde{Z}_n^{(1,2)}$. The triangle inequality yields

$$\begin{aligned}
&\left|\Pr\left(\hat{Z}_n^{(1,2)} \in C\right) - \Pr\left(\tilde{Z}_n^{(1,2)} \in C\right)\right| \\
&\leq \left|\Pr\left(\hat{Z}_n^{(1,2)} \in C\right) - \Pr\left(\hat{Z}_n^{(1,2),\Delta} \in C\right)\right| + \left|\Pr\left(\hat{Z}_n^{(1,2),\Delta} \in C\right) - \Pr\left(\tilde{Z}_n^{(1,2)} \in C\right)\right|. \qquad \text{(B.2)}
\end{aligned}$$

Examining the first term in (B.2), since $\hat{Z}_n^{(1,2)}$ and $\hat{Z}_n^{(1,2),\Delta}$ are equal conditional on $\left\{\left\|\hat{\Sigma}_n\Sigma_n^{-1} - I_p\right\|_F \leq \Delta\right\}$,

$$\begin{aligned}\left|\Pr\left(\hat{Z}_n^{(1,2)} \in C\right) - \Pr\left(\hat{Z}_n^{(1,2),\Delta} \in C\right)\right| &\leq \Pr\left(\left\|\hat{\Sigma}_n\Sigma_n^{-1} - I_p\right\|_F > \Delta\right) \\ &\leq \Pr\left(\frac{\left\|\hat{\Sigma}_n - \Sigma_n\right\|_F}{\lambda_{\min}\left(\Sigma_n\right)} > \Delta\right) \\ &\leq \Pr\left(\left\|\hat{\Sigma}_n - \Sigma_n\right\|_F > \Delta\lambda_{\min}\left(\Sigma_n\right)\right) \quad \text{(B.3)}\end{aligned}$$

since

$$\left\|\hat{\Sigma}_n\Sigma_n^{-1} - I_p\right\|_F = \left\|\left(\hat{\Sigma}_n - \Sigma_n\right)\Sigma_n^{-1}\right\|_F \leq \lambda_{\max}\left(\Sigma_n^{-1}\right)\left\|\hat{\Sigma}_n - \Sigma_n\right\|_F = \frac{\left\|\hat{\Sigma}_n - \Sigma_n\right\|_F}{\lambda_{\min}\left(\Sigma_n\right)}.$$

Examining the second term in (B.2), recall that by construction $\tilde{Z}_n^{(1,2)}$ and $\hat{Z}_n^{(1,2),\Delta}$ are equal conditional on $Q_\Delta := \left\{\left\|\hat{\Sigma}_n\Sigma_n^{-1} - I_p\right\|_F > \Delta\right\}$. Thus,

$$\begin{aligned}&\left|\Pr\left(\hat{Z}_n^{(1,2),\Delta} \in C\right) - \Pr\left(\tilde{Z}_n^{(1,2)} \in C\right)\right| \\ &= \left|\Pr\left(\hat{Z}_n^{(1,2),\Delta} \in C, Q_\Delta\right) + \Pr\left(\hat{Z}_n^{(1,2),\Delta} \in C, Q_\Delta^c\right) - \Pr\left(\tilde{Z}_n^{(1,2)} \in C, Q_\Delta\right) - \Pr\left(\tilde{Z}_n^{(1,2)} \in C, Q_\Delta^c\right)\right| \\ &= \left|\Pr\left(\tilde{Z}_n^{(1,2)} \in C, Q_\Delta\right) + \Pr\left(\hat{Z}_n^{(1,2)} \in C, Q_\Delta^c\right) - \Pr\left(\tilde{Z}_n^{(1,2)} \in C, Q_\Delta\right) - \Pr\left(\tilde{Z}_n^{(1,2)} \in C, Q_\Delta^c\right)\right| \\ &= \left|\Pr\left(\hat{Z}_n^{(1,2)} \in C, Q_\Delta^c\right) - \Pr\left(\tilde{Z}_n^{(1,2)} \in C, Q_\Delta^c\right)\right| \\ &= \left|\Pr\left(\hat{Z}_n^{(1,2)} \in C \mid Q_\Delta^c\right)\Pr\left(Q_\Delta^c\right) - \Pr\left(\tilde{Z}_n^{(1,2)} \in C \mid Q_\Delta^c\right)\Pr\left(Q_\Delta^c\right)\right| \\ &= \Pr\left(Q_\Delta^c\right)\left|\Pr\left(\hat{Z}_n^{(1,2)} \in C \mid Q_\Delta^c\right) - \Pr\left(\tilde{Z}_n^{(1,2)} \in C \mid Q_\Delta^c\right)\right| \\ &\leq \left|\Pr\left(\hat{Z}_n^{(1,2)} \in C \mid Q_\Delta^c\right) - \Pr\left(\tilde{Z}_n^{(1,2)} \in C \mid Q_\Delta^c\right)\right|. \quad \text{(B.4)}\end{aligned}$$

We now proceed to bound (B.4). $\hat{Z}_n^{(1,2)}$ and $\tilde{Z}_n^{(1,2)}$ are the same up to the addition of auxiliary noise $\hat{W}_n \sim \mathrm{N}\left(0, \hat{\Sigma}_n\right)$ versus $W_n \sim \mathrm{N}\left(0, \Sigma_n\right)$. Theorem 1.1. of Devroye et al. [2023] allows us to bound the total variation between two Gaussian random variables with the same mean but different covariances. To apply this result to $\hat{W}_n$ and W_n, we must condition on $\hat{\Sigma}_n$ so that $\hat{W}_n$ is Gaussian. Since $\hat{\Sigma}_n$ is a function of the data $D_n := \{(X_i, Y_i)\}_{i=1}^n$, it is sufficient to condition on D_n. By Theorem 1.1

and Equation 2 of Devroye et al. [2023], this yields the bound

$$d_{TV}\left(W_n \mid D_n = d, \hat{W}_n \mid D_n = d\right) = O\left(\left\|\hat{\Sigma}_n \Sigma_n^{-1} - I_p\right\|_F\right), \tag{B.5}$$

where here $\hat{\Sigma}_n$ depends on the data d; when d satisfies Q_Δ^c, this bound is $O(\Delta)$. To apply (B.5) to (B.4), we expand

$$\begin{aligned}
&\left|\Pr\left(\hat{Z}_n^{(1,2)} \in C \mid Q_\Delta^c\right) - \Pr\left(\tilde{Z}_n^{(1,2)} \in C \mid Q_\Delta^c\right)\right| \\
&\leq \sup_{A,B}\left|\Pr\left(\hat{Z}_n \in A, \hat{W}_n \in B \mid Q_\Delta^c\right) - \Pr\left(\hat{Z}_n \in A, W_n \in B \mid Q_\Delta^c\right)\right| \\
&\leq \sup_{A,B,\left(D:Q_\Delta^c\right)}\left|\Pr\left(\hat{W}_n \in B \mid \hat{Z}_n \in A, D_n \in D\right) - \Pr\left(W_n \in B \mid \hat{Z}_n \in A, D_n \in D\right)\right| \\
&\leq \sup_{B,\left(D:Q_\Delta^c\right)}\left|\Pr\left(\hat{W}_n \in B \mid D_n \in D\right) - \Pr\left(W_n \in B \mid D_n \in D\right)\right| \\
&\leq O\left(\Delta\right),
\end{aligned}$$

where the second line uses that $\hat{Z}_n^{(1,2)}$ and $\tilde{Z}_n^{(1,2)}$ are functions of $\left(\hat{Z}_n, \hat{W}_n\right)$ and $\left(\hat{Z}_n, W_n\right)$, respectively; the third line holds since we can always take D to be all datasets satisfying the event Q_Δ^c; the fourth line uses that $\hat{Z}_n$ is a deterministic function of D_n; and the final line follows from (B.5) for any data satisfying Q_Δ^c.

Combining this with (B.1), (B.2), and (B.3), for any $\Delta > 0$,

$$\begin{aligned}
\left|\Pr\left(\hat{Z}_n^{(1,2)} \in C\right) - \Pr\left(Z_n^{(1,2)} \in C\right)\right| \leq \Pr\left(\left\|\hat{\Sigma}_n - \Sigma_n\right\|_F > \Delta\lambda_{\min}\left(\Sigma_n\right)\right) \\
+ O\left(Kp^{1/4}n^{-1/2}\lambda_{\min}^{-3/2}\left(\Sigma_n\right) + \Delta\right).
\end{aligned}$$

□

B.2 Proof of Lemma 2

Proof. Our goal is to bound

$$\left|\Pr\left(\hat{Z}_n^{(2)} + r_n^B \in B_n \mid \hat{Z}_n^{(1)} + r_n^A \in A_n\right) - \Pr(Z_n^{(2)} \in B_n)\right|. \tag{B.6}$$

For ease of notation, recall $\hat{Z}_n^{(1,2)} := \begin{pmatrix} \hat{Z}_n^{(1)} \\ \hat{Z}_n^{(2)} \end{pmatrix}$, and let $C_n := A_n \times B_n$ and $r_n := \left(r_n^A, r_n^B\right)$. To bound the conditional probability in (B.6), we study the following bound on the joint probability:

$$\left|\Pr\left(\hat{Z}_n^{(1)} + r_n^A \in A_n, \hat{Z}_n^{(2)} + r_n^B \in B_n\right) - \Pr\left(Z_n^{(1)} \in A_n, Z_n^{(2)} \in B_n\right)\right| \tag{B.7}$$

$$\leq \left|\Pr\left(\hat{Z}_n^{(1,2)} + r_n \in C_n\right) - \Pr\left(\hat{Z}_n^{(1,2)} \in C_n\right)\right| \tag{B.8}$$

$$+ \left|\Pr\left(\hat{Z}_n^{(1,2)} \in C_n\right) - \Pr\left(Z_n^{(1,2)} \in C_n\right)\right|. \tag{B.9}$$

Since by statement of the Lemma, C_n can be written as the union of K almost surely disjoint, measurable convex sets, we can bound (B.9) using the Berry-Esseen-type result of Lemma 1, yielding

$$\begin{aligned}\left|\Pr\left(\hat{Z}_n^{(1,2)} \in C_n\right) - \Pr\left(Z_n^{(1,2)} \in C_n\right)\right| \leq \Pr\left(\left\|\hat{\Sigma}_n - \Sigma_n\right\|_F > \Delta\lambda_{\min}\left(\Sigma_n\right)\right) \\ + O\left(Kp^{1/4}n^{-1/2}\lambda_{\min}^{-3/2}\left(\Sigma_n\right) + \Delta\right)\end{aligned} \tag{B.10}$$

for any $\Delta > 0$.

The key technical challenge remaining is to bound (B.8). Since by assumption we can write C_n as the union of the K almost surely disjoint convex sets $\{C_{n,k}\}_{k=1}^K$, we can first upper-bound (B.8) using the triangle inequality:

$$\begin{aligned}&\left|\Pr\left(\hat{Z}_n^{(1,2)} + r_n \in C_n\right) - \Pr\left(\hat{Z}_n^{(1,2)} \in C_n\right)\right| \\ &= \left|\sum_{k=1}^K \Pr\left(\hat{Z}_n^{(1,2)} + r_n \in C_{n,k}\right) - \sum_{k=1}^K \Pr\left(\hat{Z}_n^{(1,2)} \in C_{n,k}\right)\right| \\ &\leq \sum_{k=1}^K \left|\Pr\left(\hat{Z}_n^{(1,2)} + r_n \in C_{n,k}\right) - \Pr\left(\hat{Z}_n^{(1,2)} \in C_{n,k}\right)\right|.\end{aligned} \tag{B.11}$$

To bound each of the K terms in (B.11), we will construct upper bounds for each of the two directions, positive or negative, that the term inside the absolute value could take.

Upper bounding the positive direction: We first construct an upper bound on

$$\Pr\left(\hat{Z}_n^{(1,2)} + r_n \in C_{n,k}\right) - \Pr\left(\hat{Z}_n^{(1,2)} \in C_{n,k}\right).$$

For any $\varepsilon > 0$, let $\mathbb{B}_\varepsilon$ denote an ε-ball in $\mathbb{R}^{2p}$ and let

$$C_{n,k}^{\varepsilon} := C_{n,k} + \mathbb{B}_\varepsilon = \{c + r : c \in C_{n,k}, \|r\| \leq \varepsilon\}; \tag{B.12}$$

$C_{n,k}^{\varepsilon}$ remains a convex set. We obtain the bound

$$\begin{aligned}
&\Pr\left(\hat{Z}_n^{(1,2)} + r_n \in C_{n,k}\right) - \Pr\left(\hat{Z}_n^{(1,2)} \in C_{n,k}\right) \\
&\leq \Pr\left(\hat{Z}_n^{(1,2)} + r_n \in C_{n,k}, r_n \in \mathbb{B}_\varepsilon\right) - \Pr\left(\hat{Z}_n^{(1,2)} \in C_{n,k}\right) + \Pr\left(r_n \notin \mathbb{B}_\varepsilon\right) \\
&\leq \Pr\left(\hat{Z}_n^{(1,2)} \in C_{n,k}^{\varepsilon}, r_n \in \mathbb{B}_\varepsilon\right) - \Pr\left(\hat{Z}_n^{(1,2)} \in C_{n,k}^{\varepsilon}\right) + \Pr\left(r_n \notin \mathbb{B}_\varepsilon\right) \\
&\leq \Pr\left(\hat{Z}_n^{(1,2)} \in C_{n,k}^{\varepsilon}\right) - \Pr\left(\hat{Z}_n^{(1,2)} \in C_{n,k}\right) + \Pr\left(r_n \notin \mathbb{B}_\varepsilon\right) \\
&= \Pr\left(\hat{Z}_n^{(1,2)} \in C_{n,k}^{\varepsilon}\right) - \Pr\left(Z_n^{(1,2)} \in C_{n,k}^{\varepsilon}\right) + \Pr\left(\hat{Z}_n^{(1,2)} \in C_{n,k}\right) - \Pr\left(Z_n^{(1,2)} \in C_{n,k}\right) \\
&\quad + \Pr\left(Z_n^{(1,2)} \in C_{n,k}^{\varepsilon}\right) - \Pr\left(Z_n^{(1,2)} \in C_{n,k}\right) + \Pr\left(r_n \notin \mathbb{B}_\varepsilon\right) \\
&\leq \left|\Pr\left(\hat{Z}_n^{(1,2)} \in C_{n,k}^{\varepsilon}\right) - \Pr\left(Z_n^{(1,2)} \in C_{n,k}^{\varepsilon}\right)\right| + \left|\Pr\left(\hat{Z}_n^{(1,2)} \in C_{n,k}\right) - \Pr\left(Z_n^{(1,2)} \in C_{n,k}\right)\right| \qquad \text{(B.13)} \\
&\quad + \Pr\left(Z_n^{(1,2)} \in C_{n,k}^{\varepsilon} \setminus C_{n,k}\right) + \Pr\left(r_n \notin \mathbb{B}_\varepsilon\right). \qquad \text{(B.14)}
\end{aligned}$$

Since $C_{n,k}$ and $C_{n,k}^{\varepsilon}$ are each convex sets, for any $\Delta > 0$ we can bound (B.13) using Lemma 1:

$$\begin{aligned}
&\left|\Pr\left(\hat{Z}_n^{(1,2)} \in C_{n,k}^{\varepsilon}\right) - \Pr\left(Z_n^{(1,2)} \in C_{n,k}^{\varepsilon}\right)\right| + \left|\Pr\left(\hat{Z}_n^{(1,2)} \in C_{n,k}\right) - \Pr\left(Z_n^{(1,2)} \in C_{n,k}\right)\right| \\
&= 2\Pr\left(\left\|\bar{\Sigma}_n - \Sigma_n\right\|_F > \Delta\lambda_{\min}\left(\Sigma_n\right)\right) + O\left(p^{1/4}n^{-1/2} + \Delta\right).
\end{aligned}$$

Bounding the first term in (B.14) is the problem of bounding the ε-Gaussian perimeter. We use the result of Nazarov [2003], formally stated by Chen and Fang [2015]

as follows. First, let $C_{n,k}^{-\varepsilon}$ denote the ε-shrinkage of a set:

$$C_{n,k}^{-\varepsilon} := \{c \in C_{n,k} : c + \mathbb{B}_\varepsilon \subset C_{n,k}\}; \tag{B.15}$$

this remains a convex set.

Proposition 4 (Adapted from Proposition 2.5 [Chen and Fang, 2015]). *For any convex set $C_{n,k} \in \mathbb{R}^{2p}$ and $\varepsilon_1, \varepsilon_2 \geq 0$,*

$$\Pr\left(\left(\Sigma_n^{(1,2)}\right)^{-1/2}\left(Z_n^{(1,2)} - \mu_n^{(1,2)}\right) \in C_{n,k}^{\varepsilon_1} \setminus C_{n,k}^{-\varepsilon_2}\right) \leq (2p)^{1/2}(\varepsilon_1 + \varepsilon_2).$$

We now wish to apply Proposition 4 to bound $\Pr\left(Z_n^{(1,2)} \in C_n^{\varepsilon_1} \setminus C_n^{-\varepsilon_2}\right)$ for any $\varepsilon_1, \varepsilon_2 \geq 0$, and thus must standardize $Z_n^{(1,2)}$. Let $\tilde{C}_{n,k} := \left\{\left(\Sigma_n^{(1,2)}\right)^{-1/2}\left(c - \mu_n^{(1,2)}\right) : c \in C_{n,k}\right\}$ and observe that

$$\begin{aligned}
&\left\{\left(\Sigma_n^{(1,2)}\right)^{-1/2}\left(c - \mu_n^{(1,2)}\right) : c \in C_{n,k}^{\varepsilon_1}\right\} \\
&= \left\{\left(\Sigma_n^{(1,2)}\right)^{-1/2}\left(c + r - \mu_n^{(1,2)}\right) : c \in C_{n,k}, \|r\|_2 \leq \varepsilon_1\right\} \\
&= \left\{\left(\Sigma_n^{(1,2)}\right)^{-1/2}\left(c - \mu_n^{(1,2)}\right) + \left(\Sigma_n^{(1,2)}\right)^{-1/2} r : c \in C_{n,k}, \|r\|_2 \leq \varepsilon_1\right\} \\
&\subseteq \left\{\left(\Sigma_n^{(1,2)}\right)^{-1/2}\left(c - \mu_n^{(1,2)}\right) + r : c \in C_{n,k}, \|r\|_2 \leq \varepsilon_1 \lambda_{\max}\left(\left(\Sigma_n^{(1,2)}\right)^{-1/2}\right)\right\} \\
&= \tilde{C}_{n,k}^{\varepsilon_1 \lambda_{\max}\left(\left(\Sigma_n^{(1,2)}\right)^{-1/2}\right)} \\
&= \tilde{C}_{n,k}^{\varepsilon_1 \sqrt{\lambda_{\min}\left(\Sigma_n^{(1,2)}\right)}} \\
&\subseteq \tilde{C}_{n,k}^{\varepsilon_1 \sqrt{2\lambda_{\min}(\Sigma_n)}},
\end{aligned}$$

where the second line follows from the definition of $C_{n,k}^{\varepsilon_1}$ in (B.12), and the last line follows from the form of the variance $\Sigma_n^{(1,2)}$, given in (A.3).

Similarly,

$$
\begin{aligned}
&\left\{\left(\Sigma_n^{(1,2)}\right)^{-1/2}\left(c-\mu_n^{(1,2)}\right) : c \in C_{n,k}^{-\varepsilon_2}\right\} \\
&= \left\{\left(\Sigma_n^{(1,2)}\right)^{-1/2}\left(c-\mu_n^{(1,2)}\right) : c+\mathbb{B}_{\varepsilon_2} \subset C_{n,k}\right\} \\
&= \left\{c : c+\left\{\left(\Sigma_n^{(1,2)}\right)^{-1/2} r : r \in \mathbb{B}_{\varepsilon_2}\right\} \subset \tilde{C}_{n,k}\right\} \\
&\supseteq \left\{c : c+\left\{\lambda_{\max}\left(\left(\Sigma_n^{(1,2)}\right)^{-1/2}\right) r : r \in \mathbb{B}_{\varepsilon_2}\right\} \subset \tilde{C}_{n,k}\right\} \\
&= \left\{c : c+\left\{\lambda_{\max}\left(\left(\Sigma_n^{(1,2)}\right)^{-1/2}\right) r : r \in \mathbb{B}_{\varepsilon_2}\right\} \subset \tilde{C}_{n,k}\right\} \\
&= \tilde{C}_{n,k}^{-\varepsilon_2/\lambda_{\max}\left(\left(\Sigma_n^{(1,2)}\right)^{-1/2}\right)} \\
&= \tilde{C}_{n,k}^{-\varepsilon_2/\sqrt{\lambda_{\min}\left(\Sigma_n^{(1,2)}\right)}} \\
&\supseteq \tilde{C}_{n,k}^{-\varepsilon_2/\sqrt{2\lambda_{\min}(\Sigma_n)}}.
\end{aligned}
$$

Combining this with Proposition 4, we see that for any $\varepsilon_1, \varepsilon_2 \geq 0$,

$$
\begin{aligned}
&\Pr\left(Z_n^{(1,2)} \in C_{n,k}^{\varepsilon_1} \setminus C_{n,k}^{-\varepsilon_2}\right) \\
&\leq \Pr\left(\left(\Sigma_n^{(1,2)}\right)^{-1/2}\left(Z_n^{(1,2)}-\mu_n^{(1,2)}\right) \in \tilde{C}_{n,k}^{\varepsilon_1\sqrt{2\lambda_{\min}(\Sigma_n)}} \setminus \tilde{C}_{n,k}^{-\varepsilon_2/\sqrt{2\lambda_{\min}(\Sigma_n)}}\right) \\
&\leq (2p)^{1/2}\left(\varepsilon_1\sqrt{2\lambda_{\min}\left(\Sigma_n\right)}+\varepsilon_2/\sqrt{2\lambda_{\min}\left(\Sigma_n\right)}\right) \\
&= 2p^{1/2}\left(\varepsilon_1\lambda_{\min}^{1/2}\left(\Sigma_n\right)+\varepsilon_2\lambda_{\min}^{-1/2}\left(\Sigma_n\right)\right). \qquad \text{(B.16)}
\end{aligned}
$$

This bounds the first term in (B.14) as follows:

$$
\Pr\left(Z_n^{(1,2)} \in C_{n,k}^{\varepsilon} \setminus C_{n,k}\right) \leq 2p^{1/2}\varepsilon\lambda_{\min}^{1/2}\left(\Sigma_n\right).
$$

Putting everything together, for any $\varepsilon, \Delta > 0$, we obtain the bound

$$
\begin{aligned}
&\Pr\left(\hat{Z}_n^{(1,2)}+r_n \in C_{n,k}\right)-\Pr\left(\hat{Z}_n^{(1,2)} \in C_{n,k}\right) \\
&\leq O\left(\Pr\left(\left\|\hat{\Sigma}_n-\Sigma_n\right\|_F > \Delta\lambda_{\min}\left(\Sigma_n\right)\right)+p^{1/4}n^{-1/2}\lambda_{\min}^{-3/2}\left(\Sigma_n\right)+\Delta+2p^{1/2}\varepsilon\lambda_{\min}^{1/2}\left(\Sigma_n\right)\right) \\
&\quad+\Pr\left(r_n \notin \mathbb{B}_{\varepsilon}\right), \qquad \text{(B.17)}
\end{aligned}
$$

absorbing constants into the big-O.

Upper bounding the negative direction: We now construct an upper bound in the other sign direction on

$$\Pr\left(\hat{Z}_n^{(1,2)} \in C_{n,k}\right) - \Pr\left(\hat{Z}_n^{(1,2)} + r_n \in C_{n,k}\right).$$

Let

$$\text{inrad}(C_{n,k}) := \max\left\{\varepsilon' : \exists c \in C_{n,k} \quad \text{s.t.}\, c + \mathbb{B}_{\varepsilon'} \subset C_{n,k}\right\}$$

denote the inradius of its argument set $C_{n,k}$, and let

$$\varepsilon \leq \min\left\{\cup_{k=1}^{K}\left\{\text{inrad}(C_{n,k})\right\}\right\}. \tag{B.18}$$

We now use the fact that for any $r \in \mathbb{B}_\varepsilon$ and $z \in \mathbb{R}^{2p}$,

$$\mathbb{I}\left[z \in C_{n,k}\right] - \mathbb{I}\left[z + r_n \in C_{n,k}\right] \leq \mathbb{I}\left[z \in C_{n,k} \setminus C_{n,k}^{-\|r_n\|_2}\right], \tag{B.19}$$

where $C_{n,k}^{-\|r\|_2}$ was defined in (B.15). (B.19) holds because when the right-hand side indicator of (B.19) is 0, either (i) $z \notin C_{n,k}$ and thus $\mathbb{I}\left[z \in C_{n,k}\right] = 0$ on the left-hand side, or (ii) $z \in C_{n,k}^{-\|r\|_2}$, which is nonempty since $r \leq \varepsilon$, and so $\mathbb{I}\left[z + r \in A_n\right] = 1$

on the left-hand side. This yields the bound

$$
\begin{aligned}
&\Pr\left(\hat{Z}_n^{(1,2)} \in C_{n,k}\right) - \Pr\left(\hat{Z}_n^{(1,2)} + r_n \in C_{n,k}\right) \\
&\leq \Pr\left(\hat{Z}_n^{(1,2)} \in C_{n,k}, r_n \in \mathbb{B}_\varepsilon\right) - \Pr\left(\hat{Z}_n^{(1,2)} + r_n \in C_{n,k}, r_n \in \mathbb{B}_\varepsilon\right) + \Pr\left(r_n \notin \mathbb{B}_\varepsilon\right) \\
&\qquad\qquad\qquad

We go further and note that

$$\begin{aligned}\Pr\left(r_n \notin \mathbb{B}_\varepsilon\right) &= \Pr\left(\|r_n\| > \varepsilon\right) \\ &\leq \Pr\left(\left\|r_n^A\right\|_2 > \varepsilon/\sqrt{2} \cup \left\|r_n^B\right\|_2 > \varepsilon/\sqrt{2}\right) \\ &\leq \Pr\left(\left\|r_n^A\right\|_2 > \varepsilon/\sqrt{2}\right) + \Pr\left(\left\|r_n^B\right\|_2 > \varepsilon/\sqrt{2}\right) \\ &\leq \Pr\left(\left\|r_n^A\right\|_2 > \tfrac{\varepsilon}{2}\right) + \Pr\left(\left\|r_n^B\right\|_2 > \tfrac{\varepsilon}{2}\right).\end{aligned}$$

Combining the above bound on (B.8) with the earlier derived bound (B.10) for (B.9), we obtain a final bound on the joint probability (B.7):

$$\begin{aligned}&\left|\Pr\left(\hat{Z}_n^{(1)} + r_n^A \in A_n, \hat{Z}_n^{(2)} + r_n^B \in B_n\right) - \Pr\left(Z_n^{(1)} \in A_n, Z_n^{(2)} \in B_n\right)\right| \qquad \text{(B.25)} \\ &\leq KO\left(\Pr\left(\left\|\hat{\Sigma}_n - \Sigma_n\right\|_F > \Delta\lambda_{\min}\left(\Sigma_n\right)\right) + p^{1/4}n^{-1/2}\lambda_{\min}^{-3/2}\left(\Sigma_n\right) + \Delta + p^{1/2}\varepsilon\left(\lambda_{\min}^{1/2}\left(\Sigma_n\right) + \lambda_{\min}^{-1/2}\left(\Sigma_n\right)\right)\right) \\ &\quad + K\Pr\left(2r_n^A \notin \mathbb{B}_\varepsilon\right) + K\Pr\left(2r_n^B \notin \mathbb{B}_\varepsilon\right).\end{aligned}$$

Expanding the conditional probability (B.6): We now return to bounding (B.6):

$$\left|\Pr\left(\hat{Z}_n^{(2)}+r_n^B\in B_n|\hat{Z}_n^{(1)}+r_n^A\in A_n\right)-\Pr\left(Z^{(2)}\in B_n\right)\right|$$
$$=\left|\frac{\Pr\left(\hat{Z}_n^{(2)}+r_n^B\in B_n,\hat{Z}_n^{(1)}+r_n^A\in A_n\right)}{\Pr\left(\hat{Z}_n^{(1)}+r_n^A\in A_n\right)}-\frac{\Pr\left(Z_n^{(2)}\in B_n\right)\Pr\left(\hat{Z}_n^{(1)}+r_n^A\in A_n\right)}{\Pr\left(\hat{Z}_n^{(1)}+r_n^A\in A_n\right)}\right| \tag{B.26}$$

$$\leq\left|\frac{\Pr\left(\hat{Z}_n^{(2)}+r_n^B\in B_n,\hat{Z}_n^{(1)}+r_n^A\in A_n\right)}{\Pr\left(\hat{Z}_n^{(1)}+r_n^A\in A_n\right)}-\frac{\Pr\left(Z_n^{(2)}\in B_n,Z_n^{(1)}\in A_n\right)}{\Pr\left(\hat{Z}_n^{(1)}+r_n^A\in A_n\right)}\right|$$
$$+\left|\frac{\Pr\left(Z_n^{(2)}\in B_n,Z_n^{(1)}\in A_n\right)}{\Pr\left(\hat{Z}_n^{(1)}+r_n^A\in A_n\right)}-\frac{\Pr\left(Z_n^{(2)}\in B_n\right)\Pr\left(\hat{Z}_n^{(1)}+r_n^A\in A_n\right)}{\Pr\left(\hat{Z}_n^{(1)}+r_n^A\in A_n\right)}\right| \tag{B.27}$$

$$=\frac{\left|\Pr\left(\hat{Z}_n^{(2)}+r_n^B\in B_n,\hat{Z}_n^{(1)}+r_n^A\in A_n\right)-\Pr\left(Z_n^{(2)}\in B_n,Z_n^{(1)}\in A_n\right)\right|}{\Pr\left(\hat{Z}_n^{(1)}+r_n^A\in A_n\right)}$$
$$+\frac{\left|\Pr\left(Z_n^{(2)}\in B_n\right)\Pr\left(Z_n^{(1)}\in A_n\right)-\Pr\left(Z_n^{(2)}\in B_n\right)\Pr\left(\hat{Z}_n^{(1)}+r_n^A\in A_n\right)\right|}{\Pr\left(\hat{Z}_n^{(1)}+r_n^A\in A_n\right)} \tag{B.28}$$

$$\leq\frac{\left|\Pr\left(\hat{Z}_n^{(2)}+r_n^B\in B_n,\hat{Z}_n^{(1)}+r_n^A\in A_n\right)-\Pr\left(Z_n^{(2)}\in B_n,Z_n^{(1)}\in A_n\right)\right|}{\Pr\left(\hat{Z}_n^{(1)}+r_n^A\in A_n\right)}$$
$$+\frac{\left|\Pr\left(Z_n^{(1)}\in A_n\right)-\Pr\left(\hat{Z}_n^{(1)}+r_n^A\in A_n\right)\right|}{\Pr\left(\hat{Z}_n^{(1)}+r_n^A\in A_n\right)} \tag{B.29}$$

$$\leq K\frac{O\left(\Pr\left(\left\|\hat{\Sigma}_n-\Sigma_n\right\|_F>\Delta\lambda_{\min}\left(\Sigma_n\right)\right)+p^{1/4}n^{-1/2}\lambda_{\min}^{-3/2}\left(\Sigma_n\right)+\Delta+p^{1/2}\varepsilon\left(\lambda_{\min}^{1/2}\left(\Sigma_n\right)+\lambda_{\min}^{-1/2}\left(\Sigma_n\right)\right)\right)}{\Pr\left(\hat{Z}_n^{(1)}+r_n^A\in A_n\right)}$$
$$+2K\frac{\Pr\left(\left\|r_n^A\right\|_2>\varepsilon/2\right)+\Pr\left(\left\|r_n^B\right\|_2>\varepsilon/2\right)}{\Pr\left(\hat{Z}_n^{(1)}+r_n^A\in A_n\right)}, \tag{B.30}$$

where (B.26) expands the definition of conditional probability, (B.27) applies the triangle inequality, (B.28) uses the fact that $Z_n^{(1)}$ and $Z_n^{(2)}$ are independent, (B.29) factors out $\Pr\left(Z_n^{(2)}\in B_n\right)\leq 1$, and (B.30) bounds each of the two numerator using (B.25), in the latter numerator implicitly taking $B_n=\mathbb{R}^p$.

Bounding the denominator: Lastly, we lower bound the denominator of (B.30), $\Pr\left(\hat{Z}_n^{(1)} + r_n^A \in A_n\right)$:

$$
\begin{aligned}
&\Pr\left(\hat{Z}_n^{(1)} + r_n^A \in A_n\right) \\
&\geq \Pr\left(\hat{Z}_n^{(1)} + r_n^A \in A_n, \left\|r_n^A\right\|_2 \leq \varepsilon\right) \\
&\geq \Pr\left(\hat{Z}_n^{(1)} \in A_n^{-\varepsilon}, \left\|r_n^A\right\|_2 \leq \varepsilon\right) \\
&= 1 - \Pr\left(\left\{\hat{Z}_n^{(1)} \notin A_n^{-\varepsilon}\right\} \cup \left\{\left\|r_n^A\right\|_2 > \varepsilon\right\}\right) \\
&\geq 1 - \left|\Pr\left(\hat{Z}_n^{(1)} \notin A_n^{-\varepsilon}\right) - \Pr\left(Z_n^{(1)} \notin A_n^{-\varepsilon}\right)\right| - \Pr\left(Z_n^{(1)} \notin A_n^{-\varepsilon}\right) - \Pr\left(\left\|r_n^A\right\|_2 > \varepsilon\right) \\
&= \Pr\left(Z_n^{(1)} \in A_n^{-\varepsilon}\right) - \left|\Pr\left(\hat{Z}_n^{(1)} \notin A_n^{-\varepsilon}\right) - \Pr\left(Z_n^{(1)} \notin A_n^{-\varepsilon}\right)\right| - \Pr\left(\left\|r_n^A\right\|_2 > \varepsilon\right) \\
&\geq \Pr\left(Z_n^{(1)} \in A_n^{-\varepsilon}\right) - \Pr\left(\left\|\hat{\Sigma}_n - \Sigma_n\right\|_F > \Delta\lambda_{\min}\left(\Sigma_n\right)\right) \\
&\quad - O\left(Kp^{1/4}n^{-1/2}\lambda_{\min}^{-3/2}\left(\Sigma_n\right) + \Delta\right) - \Pr\left(\left\|r_n^A\right\|_2 > \varepsilon\right), \qquad \text{(B.31)}
\end{aligned}
$$

where the last line uses the bound from Lemma 1 on the absolute difference.

A final bound on (B.6): Recalling the notation in the statement of Lemma 2, we defined the rates

$$
\begin{aligned}
E_1 &:= p^{1/4}n^{-1/2}\lambda_{\min}^{-3/2}\left(\Sigma_n\right) + \Delta + \Pr\left(\left\|\hat{\Sigma}_n - \Sigma_n\right\|_F > \Delta\lambda_{\min}\left(\Sigma_n\right)\right), \\
E_2 &:= O\left(E_1 + p^{1/2}\varepsilon\left(\lambda_{\min}^{1/2}\left(\Sigma_n\right) + \lambda_{\min}^{-1/2}\left(\Sigma_n\right)\right)\right) + \Pr\left(\left\|r_n^A\right\|_2 > \tfrac{\varepsilon}{2}\right) + \Pr\left(\left\|r_n^B\right\|_2 > \tfrac{\varepsilon}{2}\right), \\
E_3 &:= \Pr\left(Z_n^{(1)} \in A_n^{-\varepsilon}\right) - O\left(E_1\right) - \Pr\left(\left\|r_n^A\right\|_2 > \varepsilon\right).
\end{aligned}
$$

The bound in (B.30) is

$$
\left|\Pr\left(\hat{Z}_n^{(2)} + r_n^B \in B_n \middle| \hat{Z}_n^{(1)} + r_n^A \in A_n\right) - \Pr\left(Z_n^{(2)} \in B_n\right)\right| \leq \frac{KE_2}{\Pr\left(\hat{Z}_n^{(1)} + r_n^A \in A_n\right)}.
$$

The bound in (B.31) implies

$$
\Pr\left(\hat{Z}_n^{(1)} + r_n^A \in A_n\right) \geq E_3.
$$

We must take care in the case that the denominator lower bound E_3 becomes negative, a case which implies $E_3 < KE_2$. If $E_3 < KE_2$, we can trivially upper bound (B.6) by one. Thus, we arrive at our final bound

$$\left|\Pr\left(\hat{Z}_n^{(2)} + r_n^B \in B_n \middle| \hat{Z}_n^{(1)} + r_n^A \in A_n\right) - \Pr\left(Z_n^{(2)} \in B_n\right)\right| \leq \frac{KE_2}{\max\{KE_2, E_3\}}.$$

□

B.3 Proof of Lemma 4

Proof. By first-order optimality conditions, the penalized estimator $\hat{\theta}_n^{(\lambda)}$, defined in (3.1), solves

$$\sqrt{n}P_n\dot{\ell}_{\hat{\theta}_n^{(\lambda)}} + \gamma\hat{W}_n + \eta = 0 \tag{B.32}$$

for some subderivative $\eta \in \partial\rho_\lambda\left(\hat{\theta}_n^{(\lambda)}\right)$, where $\partial\rho_\lambda\left(\hat{\theta}_n^{(\lambda)}\right)$ is the sub-differential of $\rho_\lambda(\theta)$ evaluated at $\theta = \hat{\theta}_n^{(\lambda)}$. Scaling Condition 4 by $\sqrt{n}$ yields

$$\sqrt{n}P_n\dot{\ell}_{\hat{\theta}_n^{(\lambda)}} = \sqrt{n}P_n\dot{\ell}_{\check{\theta}_n^{E,*}} + \sqrt{n}H_n(\hat{\theta}_n^{(\lambda)} - \check{\theta}_n^{E,*}) + r_n^A\left(\hat{\theta}_n^{(\lambda)}\right), \tag{B.33}$$

where $r_n^A = o_p(1)$. Combining (B.33) with the optimality condition in (B.32), there exists some $\eta \in \partial\rho_\lambda\left(\hat{\theta}_n^{(\lambda)}\right)$ such that

$$\sqrt{n}P_n\dot{\ell}_{\check{\theta}_n^{E,*}} + \gamma\hat{W}_n + \sqrt{n}H_n(\hat{\theta}_n^{(\lambda)} - \check{\theta}_n^{E,*}) + r_n^A\left(\hat{\theta}_n^{(\lambda)}\right) + \eta = 0,$$

or equivalently

$$\sqrt{n}P_n\dot{\ell}_{\check{\theta}_n^{E,*}} + \gamma\hat{W}_n - H_n\sqrt{n}\check{\theta}_n^{E,*} + r_n^A\left(\hat{\theta}_n^{(\lambda)}\right) = -H_n\sqrt{n}\hat{\theta}_n^{(\lambda)} - \eta. \tag{B.34}$$

We now use (B.34) to characterize our selection event. For any E, let $\mathbb{R}_E := \{\theta \in \mathbb{R}^p : \text{supp}(\theta) = E\}$.

$$
\begin{aligned}
&\left\{\operatorname{supp}\left(\hat{\theta}_n^{(\lambda)}\right) = E\right\} \\
&= \left\{\hat{\theta}_n^{(\lambda)} \in \mathbb{R}_E^p\right\} \\
&= \left\{\sqrt{n}\hat{\theta}_n^{(\lambda)} \in \mathbb{R}_E^p\right\} \\
&= \left\{-\sqrt{n}H_n\hat{\theta}_n^{(\lambda)} \in \left\{-H_n\theta : \theta \in \mathbb{R}_E^p\right\}\right\} \\
&= \left\{-\sqrt{n}H_n\hat{\theta}_n^{(\lambda)} - \eta \in \left\{-H_n\theta - \tilde{\eta} : \theta \in \mathbb{R}_E^p, \tilde{\eta} \in \partial\rho_\lambda(\theta)\right\}\right\} \\
&= \left\{\sqrt{n}P_n\dot{\ell}_{\check{\theta}_n^{E,*}} + \gamma\hat{W}_n - H_n\sqrt{n}\check{\theta}_n^{E,*} + r_n^A\left(\hat{\theta}_n^{(\lambda)}\right) \in \left\{-H_n\theta - \tilde{\eta} : \theta \in \mathbb{R}_E^p, \tilde{\eta} \in \partial\rho_\lambda(\theta)\right\}\right\} \\
&= \left\{\sqrt{n}P_n\dot{\ell}_{\check{\theta}_n^{E,*}} + \gamma\hat{W}_n + r_n^A\left(\hat{\theta}_n^{(\lambda)}\right) \in \left\{H_n\sqrt{n}\check{\theta}_n^{E,*} - H_n\theta - \tilde{\eta} : \theta \in \mathbb{R}_E^p, \tilde{\eta} \in \partial\rho_\lambda(\theta)\right\}\right\},
\end{aligned}
$$

where the fifth equality uses Condition 2 and (B.34).

The set $\left\{H_n\theta + \tilde{\eta} : \theta \in \mathbb{R}_E^p, \tilde{\eta} \in \partial\rho_\lambda(\theta)\right\}$ is the union of finitely many disjoint measurable and convex sets by Condition 5. Thus, the set $\left\{H_n\sqrt{n}\check{\theta}_n^{E,*} - H_n\theta - \tilde{\eta} : \theta \in \mathbb{R}_E^p, \tilde{\eta} \in \partial\rho_\lambda(\theta)\right\}$ is too, as merely an affine transformation. Thus, we can characterize the selection event as

$$
\left\{\operatorname{supp}\left(\hat{\theta}_n^{(\lambda)}\right) = E\right\} = \left\{\hat{Z}_n^{(1)} + r_n^A\left(\hat{\theta}_n^{(\lambda)}\right) \in A_n^E\right\},
$$

where $\hat{Z}_n^{(1)} = \sqrt{n}P_n\dot{\ell}_{\check{\theta}_n^{E,*}} + \gamma\hat{W}_n$ by definition in (2.12), and

$$
A_n^E := \left\{H_n\sqrt{n}\check{\theta}_n^{E,*} - H_n\theta - \tilde{\eta} : \theta \in \mathbb{R}_E^p, \tilde{\eta} \in \partial\rho_\lambda(\theta)\right\} \tag{B.35}
$$

is the union of finitely many disjoint convex sets.

□

B.4 Proof of Lemma 5

Proof. Recall that $\hat{\theta}_n^E$ is the model-E minimizer defined in (3.2). Let

$$
\check{\theta}_n^E := \left(\hat{\theta}_n^E, 0_{|E|}\right) \tag{B.36}
$$

be the expanded p-dimensional sparse vector. By first-order optimality conditions, $\check{\theta}_n^E$ solves

$$\left(\sqrt{n}P_n\dot{\ell}_{\check{\theta}_n^E} - \tfrac{1}{\gamma}\hat{W}_n\right)_E = 0_{|E|}. \tag{B.37}$$

Equivalently, since this places no restriction on the E^c coordinates, we can write

$$\sqrt{n}P_n\dot{\ell}_{\check{\theta}_n^E} - \tfrac{1}{\gamma}\hat{W}_n \in 0_{|E|} \times \mathbb{R}^{|E^c|}. \tag{B.38}$$

Scaling Condition 4 by $\sqrt{n}$ yields

$$\sqrt{n}P_n\dot{\ell}_{\check{\theta}_n^E} = \sqrt{n}P_n\dot{\ell}_{\check{\theta}_n^{E,*}} + \sqrt{n}H_n(\check{\theta}_n^E - \check{\theta}_n^{E,*}) + r_n^B(\check{\theta}_n^E), \tag{B.39}$$

where $r_n^B(\check{\theta}_n^E) = o_p(1)$. Combining (B.39) with the optimality condition in (B.38),

$$\sqrt{n}P_n\dot{\ell}_{\check{\theta}_n^{E,*}} - \tfrac{1}{\gamma}\hat{W}_n + \sqrt{n}H_n(\check{\theta}_n^E - \check{\theta}_n^{E,*}) + r_n^B(\check{\theta}_n^E) \in 0_{|E|} \times \mathbb{R}^{|E^c|}. \tag{B.40}$$

We now expand the coverage event

$$\begin{aligned}
&\left\{\xi^\top\theta_{n,j}^{E,*} \in \mathrm{CI}_n^{\xi,\alpha}\left(\hat{\theta}_n^E, \hat{S}_n^E\right)\right\} \\
&= \left\{\xi^\top\theta_n^{E,*} \in \left[\xi^\top\hat{\theta}_n^E \pm n^{-1/2}z_{1-\alpha/2}\left(\xi^\top\hat{S}_n^E\xi\right)^{1/2}\right]\right\} \\
&= \left\{\sqrt{n}\xi^\top\left(\theta_n^{E,*} - \hat{\theta}_n^E\right) \in \left[\pm z_{1-\alpha/2}\left(\xi^\top\hat{S}_n^E\xi\right)^{1/2}\right]\right\} \\
&= \left\{\sqrt{n}\left(\theta_n^{E,*} - \hat{\theta}_n^E\right) \in \left\{v \in \mathbb{R}^{|E|} : \xi^\top v \in \left[\pm z_{1-\alpha/2}\left(\xi^\top\hat{S}_n^E\xi\right)^{1/2}\right]\right\}\right\} \\
&= \left\{\sqrt{n}\left(\check{\theta}_n^{E,*} - \check{\theta}_n^E\right) \in \left\{v \in \mathbb{R}^p : \xi^\top v_E \in \left[\pm z_{1-\alpha/2}\left(\xi^\top\hat{S}_n^E\xi\right)^{1/2}\right]\right\}\right\} \\
&= \left\{\sqrt{n}H_n\left(\check{\theta}_n^{E,*} - \check{\theta}_n^E\right) \in \left\{v \in \mathbb{R}^p : \xi^\top\left(H_{n,E,E}\right)^{-1}v_E \in \left[\pm z_{1-\alpha/2}\left(\xi^\top\hat{S}_n^E\xi\right)^{1/2}\right]\right\}\right\} \\
&= \left\{\sqrt{n}P_n\dot{\ell}_{\theta_n^{E,*}} - \tfrac{1}{\gamma}\hat{W}_n + r_n^B\left(\check{\theta}_n^E\right) \in \left\{v \in \mathbb{R}^p : \xi^\top\left(H_{n,E,E}\right)^{-1}v_E \in \left[\pm z_{1-\alpha/2}\left(\xi^\top\hat{S}_n^E\xi\right)^{1/2}\right]\right\}\right\} \\
&= \left\{\hat{Z}_n^{(2)} + r_n^B\left(\check{\theta}_n^E\right) \in B_n^E\right\},
\end{aligned} \tag{B.41}$$

where the fourth equality makes use of the definitions of $\check{\theta}_n^{E,*}$ in (2.7) and $\check{\theta}_n^E$ in

(B.36), $\hat{Z}_n^{(2)} = \sqrt{n} P_n \dot{\ell}_{\check{\theta}_n^{E,*}} - \frac{1}{\gamma}\hat{W}_n$ by definition in (2.12), and

$$B_n^E := \left\{ v \in \mathbb{R}^p : \xi^\top \left(H_{n,E,E}\right)^{-1} v_E \in \left[\pm z_{1-\alpha/2} \left(\xi^\top \hat{S}_n^E \xi \right)^{1/2} \right] \right\}. \tag{B.42}$$

Note that B_n^E is a convex set since it the pre-image of a convex set with respect to an affine function. □

B.5 Proof of Theorem 6

Proof. Recalling the definition of B_n^E in (B.42), we lower bound the conditional probability

$$\begin{aligned}
&\Pr\left(\xi^\top \theta_n^{E,*} \in \mathrm{CI}_n^{\xi,\alpha}\left(\hat{\theta}_n^E, \hat{S}_n^E\right) \middle| \operatorname{supp}\left(\hat{\theta}_n^{(\lambda)}\right) = E\right) \\
&= \Pr\left(\xi^\top \theta_n^{E,*} \in \mathrm{CI}_n^{\xi,\alpha}\left(\hat{\theta}_n^E, \hat{S}_n^E\right) \middle| \operatorname{supp}\left(\hat{\theta}_n^{(\lambda)}\right) = E\right) - \Pr\left(Z_n^{(2)} \in B_n^E\right) \\
&\quad + \Pr\left(Z_n^{(2)} \in B_n^E\right) - \Pr\left(\hat{Z}_n^{(2)} + r_n^B\left(\check{\theta}_n^E\right) \in B_n^E\right) + \Pr\left(\hat{Z}_n^{(2)} + r_n^B\left(\check{\theta}_n^E\right) \in B_n^E\right) \\
&\geq \Pr\left(\hat{Z}_n^{(2)} + r_n^B\left(\check{\theta}_n^E\right) \in B_n^E\right) - \left|\Pr\left(Z_n^{(2)} \in B_n^E\right) - \Pr\left(\hat{Z}_n^{(2)} + r_n^B\left(\check{\theta}_n^E\right) \in B_n^E\right)\right| \qquad \text{(B.43)} \\
&\quad - \left|\Pr\left(\xi^\top \theta_n^{E,*} \in \mathrm{CI}_n^{\xi,\alpha}\left(\hat{\theta}_n^E, \hat{S}_n^E\right) \middle| \operatorname{supp}\left(\hat{\theta}_n^{(\lambda)}\right) = E\right) - \Pr\left(Z_n^{(2)} \in B_n^E\right)\right|. \qquad \text{(B.44)}
\end{aligned}$$

By Lemma 5, as in (B.41), and Condition 3,

$$\lim_{n\to\infty} \Pr\left(\hat{Z}_n^{(2)} + r_n^B\left(\check{\theta}_n^E\right) \in B_n^E\right) = \lim_{n\to\infty} \Pr\left(\xi^\top \theta_{n,j}^{E,*} \in \mathrm{CI}_n^{\xi,\alpha}\left(\hat{\theta}_n^E, \hat{S}_n^E\right)\right) \geq 1 - \alpha.$$

It remains to show that each of the remaining terms in (B.43) and (B.44) are $o(1)$.

Bounding (B.44) We apply Lemma 2 to the sets A_n^E introduced in Lemma 4, and B_n^E introduced in Lemma 5. For clarity, we begin by restating Lemma 2 using the notation of Lemma 4 and Lemma 5.

Since A_n^E (defined in (B.35)) is the union of $2^{|E|}$ disjoint convex sets, and B_n^E (defined in (B.42)) is a convex set, $A_n^E \times B_n^E$ can be written as $K = 2^{|E|}$ convex sets. Let $\Delta > 0$, and $\varepsilon > 0$ be less than the smallest inradii of these K sets, defined

in (B.18). For brevity, define the terms

$$E_1 := p^{1/4} n^{-1/2} \lambda_{\min}^{-3/2}(\Sigma_n) + \Delta + \Pr\left(\left\|\hat{\Sigma}_n - \Sigma_n\right\|_F > \Delta \lambda_{\min}(\Sigma_n)\right), \tag{B.45}$$

$$E_2 := O\left(E_1 + p^{1/2}\varepsilon\left(\lambda_{\min}^{1/2}(\Sigma_n) + \lambda_{\min}^{-1/2}(\Sigma_n)\right)\right) + \Pr\left(\left\|r_n^A\right\|_2 > \tfrac{\varepsilon}{2}\right) + \Pr\left(\left\|r_n^B\right\|_2 > \tfrac{\varepsilon}{2}\right), \tag{B.46}$$

$$E_3 := \Pr\left(Z_n^{(1)} \in A_n^{-\varepsilon}\right) - O(E_1) - \Pr\left(\left\|r_n^A\right\|_2 > \varepsilon\right), \tag{B.47}$$

where $r_n^A\left(\hat{\theta}_n^{(\lambda)}\right)$ defined in (B.33), $r_n^B\left(\check{\theta}_n^E\right)$ defined in (B.39), and $\left(A_n^E\right)^{-\varepsilon}$ is the ε-shrinkage (defined in (B.15)) of A_n^E. If Condition 1 holds, then

$$\left|\Pr\left(\hat{Z}_n^{(2)} + r_n^B\left(\check{\theta}_n^E\right) \in B_n^E \middle| \hat{Z}_n^{(1)} + r_n^A\left(\hat{\theta}_n^{(\lambda)}\right) \in A_n^E\right) - \Pr\left(Z_n^{(2)} \in B_n^E\right)\right| \leq \frac{KE_2}{\max\{E_3, KE_2\}}. \tag{B.48}$$

It remains to show that the bound vanishes. First, we establish that for large enough n the inradii are bounded above zero and hence we can fix an $\varepsilon > 0$. To show this, it suffices to show that B_n^E converges to a fixed set, and that each of the 2^K constituent sets of A_n^E converge to fixed sets with non-empty interiors.

To begin, recall from (B.42) that

$$B_n^E := \left\{v \in \mathbb{R}^p : \xi^\top \left(H_{n,E,E}\right)^{-1} v_E \in \left[\pm z_{1-\alpha/2}\left(\xi^\top \hat{S}_n^E \xi\right)^{1/2}\right]\right\}.$$

By Assumption 1, $H_{n,E,E} \to H_{E,E}$ and $S_n^E \to S^E := \left(\hat{H}_{E,E}\right)^{-1} \hat{\Sigma}_{E,E} \left(\hat{H}_{E,E}\right)^{-1}$, and thus

$$B_n^E \to B^E := \left\{v \in \mathbb{R}^p : \xi^\top \left(H_{E,E}\right)^{-1} v_E \in \left[\pm z_{1-\alpha/2}\left(\xi^\top S^E \xi\right)^{1/2}\right]\right\}.$$

In the E^c coordinates, the elements of B^E are unrestricted. By Assumption 1, H and Σ are positive definite and so $\left[\pm z_{1-\alpha/2}\left((1+\frac{1}{\gamma^2})\xi^\top S^E \xi\right)^{1/2}\right]$ contains a non-empty open set. Since the maximum eigenvalue of $\left(H_{E,E}\right)^{-1}$ is also bounded away from zero by Assumption 1, the elements of B^E contain a non-empty open set in the E coordinates with inradius bounded away from zero. Thus, B^E contains a

non-empty open set.

Now, recall from (B.35) that

$$A_n^E := \left\{ H_n \sqrt{n} \check{\theta}_n^{E,*} - H_n \theta - \tilde{\eta} : \theta \in \mathbb{R}_E^p, \tilde{\eta} \in \partial \rho_\lambda(\theta) \right\}.$$

By Assumption 1(i) and 1(iii),

$$A_n^E \to A^E := \left\{ H \theta^{E,*} - H\theta - \tilde{\eta} : \theta \in \mathbb{R}_E^p, \tilde{\eta} \in \partial \rho_\lambda(\theta) \right\},$$

which by Condition 5 can be written as the union of finitely many disjoint convex sets with non-empty interiors.

It remains to show that the upper bound in (B.48) asymptotically vanishes. We examine each of the three constituent terms E_1, E_2, and E_3. First, recall from (B.45) that

$$E_1 := p^{1/4} n^{-1/2} \lambda_{\min}^{-3/2}(\Sigma_n) + \Delta + \Pr\left(\left\| \hat{\Sigma}_n - \Sigma_n \right\|_F > \Delta \lambda_{\min}(\Sigma_n) \right).$$

Since $\lambda_{\min}^{-3/2}(\Sigma_n) \to \lambda_{\min}^{-3/2}(\Sigma) < \infty$ by Assumption 1, clearly $p^{1/4} n^{-1/2} \lambda_{\min}^{-3/2}(\Sigma_n) \to 0$. Similarly, since $\lambda_{\min}(\Sigma_n) \to \lambda_{\min}(\Sigma) > 0$ by Assumption 1, by Condition 6 and Markov's inequality, for any $\Delta > 0$

$$\Pr\left(\left\| \hat{\Sigma}_n - \Sigma_n \right\|_F > \Delta \lambda_{\min}(\Sigma_n) \right) = o(1).$$

Since Δ was arbitrary, we can choose a sequence Δ_n such that

$$\Delta_n + \Pr\left(\left\| \hat{\Sigma}_n - \Sigma_n \right\|_F > \Delta_n \lambda_{\min}(\Sigma_n) \right) = o(1).$$

Now recall from (B.46) that

$$E_2 := O\left(E_1 + p^{1/2} \varepsilon \left(\lambda_{\min}^{1/2}(\Sigma_n) + \lambda_{\min}^{-1/2}(\Sigma_n) \right) \right) + \Pr\left(\left\| r_n^A \right\|_2 > \tfrac{\varepsilon}{2} \right) + \Pr\left(\left\| r_n^B \right\|_2 > \tfrac{\varepsilon}{2} \right).$$

By Lemma 4 and Lemma 5, for any $\varepsilon > 0$,

$$\Pr\left(\left\|r_n^A\left(\hat{\theta}_n^{(\lambda)}\right)\right\|_2 > \tfrac{\varepsilon}{2}\right) + \Pr\left(\left\|r_n^B\left(\check{\theta}_n^E\right)\right\|_2 > \tfrac{\varepsilon}{2}\right) = o(1).$$

Since ε was arbitrarily small and $\lambda_{\min}(\Sigma_n) \to \lambda_{\min}(\Sigma)$ by Assumption 1, $\varepsilon\left(\lambda_{\min}^{1/2}(\Sigma_n) + \lambda_{\min}^{-1/2}(\Sigma_n)\right) \to 0$ and so $E_2 \to 0$.

Lastly, recall from (B.47) that

$$E_3 := \Pr\left(Z_n^{(1)} \in A_n^{-\varepsilon}\right) - O(E_1) - \Pr\left(\left\|r_n^A\right\|_2 > \varepsilon\right).$$

By the logic used earlier to analyze E_2, $O(E_1) - \Pr\left(\left\|r_n^A\right\|_2 > \varepsilon\right) \to 0$, and so it remains to study $\Pr\left(Z_n^{(1)} \in \left(A_n^E\right)^{-\varepsilon}\right)$. Since ε is less than the inradius of the limit A^E of A_n^E and Σ_n converges to the positive definite Σ. It follows that

$$\liminf_{n\to\infty} \Pr\left(Z_n^{(1)} \in \left(A_n^E\right)^{-\varepsilon}\right) > 0.$$

This establishes that the upper bound in (B.48) asymptotically vanishes:

$$\lim_{n\to\infty} \frac{E_2}{\max\{E_2, E_3\}} = 0.$$

Bounding (B.43) We need to show that

$$\lim_{n\to\infty} \left|\Pr\left(\hat{Z}_n^{(2)} + r_n^B\left(\check{\theta}_n^E\right) \in B_n^E\right) - \Pr\left(Z_n^{(2)} \in B_n^E\right)\right| = 0.$$

(B.25) upper bounded this absolute difference with E_2 defined in (B.46). As previously established, $\lim_{n\to\infty} E_2 = 0$.

□

B.6 Proof of Corollary 7

Under Condition 7, it is clear to see that

$$\ell_\theta\left(Y_i - \tfrac{1}{\gamma}\bar{W}_{n,i}\right) = \ell_\theta(Y_i) - \tfrac{1}{\gamma}\theta^\top A_i \bar{W}_{n,i},$$

where $A_i\bar{W}_{n,i} \sim \mathrm{N}\left(0, A_i\bar{\Sigma}_{n,i}A_i^\top\right)$. Thus, analyzing $\bar{\theta}_n^E$,

$$\begin{aligned}
\bar{\theta}_n^E &:= \underset{\theta\in\mathbb{R}^{|E|}}{\arg\min}\left\{\frac{1}{n}\sum_{i=1}^n \ell_{(\theta,0_{p-|E|})}\left(Y_i - \tfrac{1}{\gamma}\bar{W}_{n,i}\right)\right\} \\
&= \underset{\theta\in\mathbb{R}^{|E|}}{\arg\min}\left\{\frac{1}{n}\sum_{i=1}^n \left[\ell_{(\theta,0_{p-|E|})}(Y_i) - \tfrac{1}{\gamma}\left(\theta, 0_{p-|E|}\right)^\top A_i\bar{W}_{n,i}\right]\right\} \\
&= \underset{\theta\in\mathbb{R}^{|E|}}{\arg\min}\left\{\frac{1}{n}\sum_{i=1}^n \ell_{(\theta,0_{p-|E|})}(Y_i) - \left(\theta, 0_{p-|E|}\right)^\top \frac{1}{n}\sum_{i=1}^n \tfrac{1}{\gamma}A_i\bar{W}_{n,i}\right\} \\
&= \underset{\theta\in\mathbb{R}^{|E|}}{\arg\min}\left\{P_n\ell_{(\theta,0_{p-|E|})} - \tfrac{1}{\gamma}n^{-1/2}\left(\theta, 0_{p-|E|}\right)^\top \hat{W}_n\right\} \\
&= \underset{\theta\in\mathbb{R}^{|E|}}{\arg\min}\left\{P_n\ell_{(\theta,0_{p-|E|})} - \tfrac{1}{\gamma}n^{-1/2}\theta^\top\hat{W}_{n,E}\right\} =: \hat{\theta}_n^E,
\end{aligned}$$

if $\hat{W}_n = n^{-1/2}\sum_{i=1}^n A_i\bar{W}_{n,i}$. A similar proof shows that $\hat{\theta}_n^{(\lambda)} = \bar{\theta}_n^{(\lambda)}$. By equality, we can apply Theorem 6.

B.7 Proof of Proposition 8

Proof. We first establish that $\hat{\theta}_n^{(\lambda)}$, defined in (3.1), converges to the unique (by assumption) population quantity

$$\theta_n^* := \underset{\theta\in\Theta}{\arg\min}\,\mathbb{E}P_n\ell_\theta, \tag{B.49}$$

and that we have consistency of $\hat{\theta}_n^E \xrightarrow{\mathrm{p}} \theta_n^{E,*}$, where these two quantities were defined in (3.2) and (2.1).

By Theorem 5.7 of van der Vaart [1998], since (i) our estimands are unique, (ii) we have uniform convergence, and $\hat{\theta}_n^{(\lambda)}$ is a near minimizer of $\mathbb{E}P_n\ell_\theta$ (as the penalty and randomized terms are $o_p(1)$), it follows that $\hat{\theta}_n^E \xrightarrow{\mathrm{p}} \theta_n^{E,*}$ and $\hat{\theta}_n^{(\lambda)} \xrightarrow{\mathrm{p}} \theta_n^*$.

Having established consistency, we are ready to show that Condition 4 is met. The following results and conditions are standard (see van der Vaart [1998, Theorem 5.41]). By existence of third derivatives, for any θ we can construct the following

Taylor expansion of $P_n\dot{\ell}_\theta$ around $\check{\theta}_n^{E,*}$ defined in (2.7):

$$P_n\dot{\ell}_\theta = P_n\dot{\ell}_{\check{\theta}_n^{E,*}} + \left[P_n\ddot{\ell}_{\check{\theta}_n^{E,*}}\right](\theta - \check{\theta}_n^{E,*}) + \tfrac{1}{2}(\theta - \check{\theta}_n^{E,*})^\top \left[P_n\dddot{\ell}_{\tilde{\theta}_n}\right](\theta - \check{\theta}_n^{E,*}) \quad \text{(B.50)}$$

$$= P_n\dot{\ell}_{\check{\theta}_n^{E,*}} + H_n(\theta - \check{\theta}_n^{E,*}) + r_n(\theta), \quad \text{(B.51)}$$

where

$$r_n(\theta) := \left(P_n\ddot{\ell}_{\check{\theta}_n^{E,*}} - H_n + \tfrac{1}{2}(\theta - \check{\theta}_n^{E,*})^\top \left[P_n\dddot{\ell}_{\tilde{\theta}_n}\right]\right)(\theta - \check{\theta}_n^{E,*}) \quad \text{(B.52)}$$

for some $\tilde{\theta}_n$ which lies on the line segment between θ and $\check{\theta}_n^{E,*}$.

We now characterize terms in the above expressions. By the law of large numbers, $P_n\ddot{\ell}_{\check{\theta}_n^{E,*}} - H_n = o_p(1)$ where $H_n := \mathbb{E}P_n\ddot{\ell}_{\check{\theta}_n^{E,*}}$. Lastly, provided that $\dddot{\ell}_{\tilde{\theta}_n}(Y_i)$ is dominated by some integrable function $\dddot{\ell}(Y_i)$ for any $\tilde{\theta}_n$ in some neighborhood $\mathcal{N}_n$,

$$\sup_{\tilde{\theta}_n \in \mathcal{N}_n} \left\| P_n\dddot{\ell}_{\tilde{\theta}_n} \right\|_F \leq \left\| P_n\dddot{\ell} \right\|_F = O_p(1) \quad \text{(B.53)}$$

by the law of large numbers. (B.53) will clearly hold if such a $\dddot{\ell}_{\tilde{\theta}_n}(Y_i)$ exists for all $\tilde{\theta}_n \in \Theta$. Alternatively, (B.53) will hold with probability tending to one if $\mathcal{N}_n$ is an appropriate neighborhood around θ_n^* in the case of a Taylor expansion of $P_n\dot{\ell}_{\hat{\theta}_n^{(\lambda)}}$, or neighborhood around $\check{\theta}_n^{E,*}$ in the case of $P_n\dot{\ell}_{\check{\theta}_n^E}$.

Analyzing the inference estimator $\check{\theta}_n^E$: The first order optimality condition of (3.2) implies that

$$\sqrt{n}\left(P_n\dot{\ell}_{\check{\theta}_n^E} - \tfrac{1}{\gamma}\hat{W}_n\right)_E = 0.$$

Replacing $P_n\dot{\ell}_{\check{\theta}_n^E}$ with its Taylor expansion in (B.51), we arrive at the classical results $\sqrt{n}\left(\hat{\theta}_n^E - \theta_n^{E,*}\right) = O_p(1)$. This turn implies that $\check{\theta}_n^E - \check{\theta}_n^{E,*} = O_p\left(n^{-1/2}\right)$, and furthermore that

$$r_n\left(\check{\theta}_n^E\right) := \left(P_n\ddot{\ell}_{\check{\theta}_n^{E,*}} - H_n + \tfrac{1}{2}(\check{\theta}_n^E - \check{\theta}_n^{E,*})^\top \left[P_n\dddot{\ell}_{\tilde{\theta}}\right]\right)(\check{\theta}_n^E - \check{\theta}_n^{E,*}) = o_p\left(n^{-1/2}\right).$$

Analyzing the selection estimator $\hat{\theta}_n^{(\lambda)}$: We now turn to analyzing $r_n\left(\hat{\theta}_n^{(\lambda)}\right)$. By the first order optimality condition of (3.1),

$$\sqrt{n}P_n\dot{\ell}_{\hat{\theta}_n^{(\lambda)}} + \gamma\hat{W}_n + \eta = 0,$$

for some $\eta \in \partial\rho_\lambda\left(\hat{\theta}_n^{(\lambda)}\right)$.

As a first step, show that under a local asymptotic setting, $\sqrt{n}\hat{\theta}_n^{(\lambda)} = O_p(1)$. We replace $P_n\dot{\ell}_{\hat{\theta}_n^{(\lambda)}}$ with its Taylor expansion around θ_n^* in (B.49):

$$\sqrt{n}P_n\dot{\ell}_{\theta_n^*} + \gamma\hat{W}_n + \eta + \sqrt{n}(\hat{\theta}_n^{(\lambda)} - \theta_n^*)\left[P_n\ddot{\ell}_{\theta_n^*} + \tfrac{1}{2}(\hat{\theta}_n^{(\lambda)} - \theta_n^*)^\top\left[P_n\dddot{\ell}_{\tilde{\theta}'_n}\right]\right] = 0$$

for some $\tilde{\theta}'_n$ between $\hat{\theta}_n^{(\lambda)}$ and θ_n^*. Since $\hat{W}_n = O_p(1)$ and η is bounded, under typical regularity conditions $\sqrt{n}P_n\dot{\ell}_{\theta_n^*} = O_p(1)$ and it follows that $\sqrt{n}(\hat{\theta}_n^{(\lambda)} - \theta_n^*) = O_p(1)$. Our assumed local alternative setting where $\sqrt{n}\theta_n^* = O(1)$ implies that $\sqrt{n}\hat{\theta}_n^{(\lambda)} = O_p(1)$.

We now demonstrate that $r_n\left(\hat{\theta}_n^{(\lambda)}\right) = o_p\left(n^{-1/2}\right)$. We replace $P_n\dot{\ell}_{\hat{\theta}_n^{(\lambda)}}$ with its Taylor expansion in (B.50):

$$\sqrt{n}P_n\dot{\ell}_{\check{\theta}_n^{E,*}} + \gamma\hat{W}_n + \eta + \sqrt{n}(\hat{\theta}_n^{(\lambda)} - \check{\theta}_n^{E,*})\left[P_n\ddot{\ell}_{\check{\theta}_n^{E,*}} + \tfrac{1}{2}(\hat{\theta}_n^{(\lambda)} - \check{\theta}_n^{E,*})^\top\left[P_n\dddot{\ell}_{\tilde{\theta}_n}\right]\right] = 0$$

for some $\tilde{\theta}_n$ between $\hat{\theta}_n^{(\lambda)}$ and $\check{\theta}_n^{E,*}$. By Assumption 1[i], we know that $\sqrt{n}\check{\theta}_n^{E,*} = O(1)$ and thus that $\sqrt{n}(\hat{\theta}_n^{(\lambda)} - \check{\theta}_n^{E,*}) = O_p(1)$. This in turn implies that

$$\sqrt{n}r_n\left(\hat{\theta}_n^{(\lambda)}\right) := \sqrt{n}(\hat{\theta}_n^{(\lambda)} - \check{\theta}_n^{E,*})\left(P_n\ddot{\ell}_{\check{\theta}_n^{E,*}} - H_n + \tfrac{1}{2}(\hat{\theta}_n^{(\lambda)} - \check{\theta}_n^{E,*})^\top\left[P_n\dddot{\ell}_{\tilde{\theta}_n}\right]\right) = o_p(1)$$

as desired, since $P_n\ddot{\ell}_{\check{\theta}_n^{E,*}} - H_n = o_p(1)$ by the law of large numbers and $P_n\dddot{\ell}_{\tilde{\theta}_n} = O_p(1)$.

□

B.8 Proof of Proposition 3

Proof. Recall that for $\rho_\lambda(\theta) := \lambda \|\theta\|_1$, for any $j \in [p]$,

$$[\partial \rho_\lambda(\theta)]_j := \begin{cases} \lambda \operatorname{sign}(\theta_j) & \text{if } \theta_j \neq 0 \\ [-\lambda, \lambda] & \text{if } \theta_j = 0 \end{cases}.$$

For a fixed λ, this is clearly bounded and thus satisfies Condition 5(i). It remains to show that Condition 5(ii) holds, i.e., for any positive definite matrix $H \in \mathbb{R}^{p \times p}$,

$$\{H\theta + \eta : \operatorname{supp}(\theta) = E, \eta \in \partial \rho_\lambda(\theta)\} \tag{B.54}$$

has a non-empty interior and can be written as the union of finitely many disjoint convex sets.

To begin, let $s_E \in \{-1, 1\}^{|E|}$ be one of the $2^{|E|}$ possible combinations of signs of θ_E, $\mathbb{R}^p_{E,s_E} := \{\theta : \operatorname{supp}(\theta) = E, \operatorname{sign}(\theta_E) = s_E\}$ be an orthant, and $\mathcal{E}_{E,s_E} := \left\{\eta : \eta \in \partial \rho_\lambda(\theta), \theta \in \mathbb{R}^p_{E,s_E}\right\}$ be a collection of subgradients. We can then write

$$\begin{aligned} &\{H\theta + \eta : \operatorname{supp}(\theta) = E, \eta \in \partial \rho_\lambda(\theta)\} \\ &= \bigcup_{s_E \in \{-1,1\}^{|E|}} \left\{H\theta + \eta : \theta \in \mathbb{R}^p_{E,s_E}, \eta \in \partial \rho_\lambda(\theta)\right\} \\ &= \bigcup_{s_E \in \{-1,1\}^{|E|}} \left\{H\theta : \theta \in \mathbb{R}^p_{E,s_E}\right\} + \mathcal{E}_{E,s_E}, \end{aligned}$$

where the final sum denotes the Minkowski sum, i.e., the pairwise sum of all points in the two sets.

First, we can see that $\left\{H\theta : \theta \in \mathbb{R}^p_{E,s_E}\right\}$ is an affine transformation of a convex set, $\mathcal{E}_{E,s_E}$ is a convex set, and so their Minkowski sum is a convex set. The union encompasses $2^{|E|}$ such convex sets. To show that these sets are disjoint, for sake of contradiction assume that there is a point lying in two such sets. This implies that there exists some $s_E \neq s'_E \in \{-1, 1\}^{|E|}$ and $\theta \in \mathbb{R}^p_{E,s_E}, \theta' \in \mathbb{R}^p_{E,s'_E}$ such that

$H_{E,E}\theta_E + \lambda s_E = H_{E,E}\theta'_E + \lambda s'_E$. Rearranging,

$$H_{E,E}(\theta_E - \theta'_E) = \lambda(s'_E - s_E).$$

Since H is positive definite, $(\theta_E - \theta'_E)^\top H_{E,E}(\theta_E - \theta'_E) > 0$. However, $\lambda(\theta_E - \theta'_E)^\top(s'_E - s_E) < 0$ by definition of the signs. Thus, we have a contradiction and so (B.54) is the union of $2^{|E|}$ disjoint convex sets.

It remains to show that these sets have non-empty interiors, for which it suffices to show that $\left\{H\theta : \theta \in \mathbb{R}^p_{E,s_E}\right\} + \mathcal{E}_{E,s_E}$ contains a non-empty open set. In the E coordinates of $\left\{H\theta : \theta \in \mathbb{R}^p_{E,s_E}\right\}$, since $\mathbb{R}^p_{E,s_E}$ contains a non-empty open set, and H is positive definite, $\{(H\theta)_E : \text{supp}\,(\theta) = E, \text{sign}(\theta_E) = s_E\}$ also contains a non-empty open set. In the E^c coordinates, we can clearly see that $\mathcal{E}_{E,s_E}$ contains a non-empty open set and thus the Minkowski sum $\left\{(H\theta) : \theta \in \mathbb{R}^p_{E,s_E}\right\} + \mathcal{E}_{E,s_E}$ does too. □

C Supplemental results

C.1 An algorithm implementing Theorem 6 for L_1-penalized GLMs

Here, we additionally provide the procedure for obtaining valid inference using Algorithm Theorem 6.

Algorithm 2 is quite similar to Algorithm 1. A minor but key differences lies in the sandwich variance estimator $\hat{S}^E_n$ in Step 5. Algorithm 1 computes the "meat" of the sandwich using squared residuals relative to the noisy outcomes $\left\{Y_i - \frac{1}{\gamma}\tilde{W}_{n,i}\right\}_{i=1}^n$; this is the classical approach in the M-estimation literature [White, 1980]. However, Theorem 6 does not use noisy outcomes and so there is not an immediately natural estimator $\hat{S}^E_n$ available satisfying Condition 3. As a solution, we use the estimator in (C.1) which involves squared residuals relative to the observed outcomes $\{Y_i\}_{i=1}^n$, and then additionally factor in the variance of the noise $\hat{W}_n$. The justification is as follows.

Algorithm 2: implementing Theorem 6 for L_1-penalized GLMs

Input: Data $\{Y_i, X_i\}$, penalty λ, information split γ.

1. $\tilde{\theta}_n \leftarrow \arg\min_{\theta \in \mathbb{R}^p} P_n \ell_\theta$.
2. Sample noise $\hat{W}_n \leftarrow \mathrm{N}\left(0, \frac{1}{n}\sum_{i=1}^n \hat{\alpha}\left(\tilde{\theta}_n\right) \ddot{b}\left(X_i^\top \tilde{\theta}_n\right) X_i X_i^\top\right)$.
3. Estimate $\hat{\theta}_n^{(\lambda)}$ as in (3.1) and define $E := \mathrm{supp}\left(\hat{\theta}_n^{(\lambda)}\right)$.
4. Estimate $\hat{\theta}_n^E$ as in (3.2).
5. Compute the modified sandwich variance estimator:

$$\hat{S}_n^E \leftarrow \left(H_{n,E,E}\left(\hat{\theta}_n^E\right)\right)^{-1} \tilde{\Sigma}_{n,E,E} \left(H_{n,E,E}\left(\hat{\theta}_n^E\right)\right)^{-1}$$

where

$$\tilde{\Sigma}_{n,E,E} \leftarrow \frac{1}{\gamma^2}\mathrm{Var}\left(\hat{W}_{n,E,E}\right) + \frac{1}{n}\sum_{i=1}^n \left(Y_i - \dot{b}(X_{i,E}^\top \hat{\theta}_n^E)\right)^2 X_{i,E} X_{i,E}^\top. \tag{C.1}$$

Output: A confidence interval centered at $\hat{\theta}_n^E$,

$$\mathrm{CI}_n^{\xi,\alpha}\left(\hat{\theta}_n^E, \hat{S}_n^E\right) = \left[\xi^\top \hat{\theta}_n^E \pm n^{-1/2} z_{1-\alpha/2}\left(\xi^\top \hat{S}_n^E \xi\right)^{1/2}\right].$$

In the setting of Corollary 7, the canonical sandwich estimator in Algorithm 1 takes the form

$$\frac{1}{n}\sum_{i=1}^n \left(Y_i - \frac{1}{\gamma}\tilde{W}_{n,i} - \dot{b}(X_{i,E}^\top \hat{\theta}_n^E)\right)^2 X_{i,E} X_{i,E}^\top.$$

Expanding the quadratic, this equals

$$\frac{1}{n}\sum_{i=1}^n \left[\left(Y_i - \dot{b}(X_{i,E}^\top \hat{\theta}_n^E)\right)^2 - 2\frac{1}{\gamma}\tilde{W}_{n,i} Y_i + 2\frac{1}{\gamma}\tilde{W}_{n,i}\dot{b}(X_{i,E}^\top \hat{\theta}_n^E) + \frac{1}{\gamma^2}\tilde{W}_{n,i}^2\right] X_{i,E} X_{i,E}^\top.$$

Examining the terms in the quadratic expansion, note that

$$\mathbb{E}\left[\tilde{W}_{n,i} Y_i\right] = \mathbb{E}\left[\tilde{W}_{n,i}\dot{b}(X_{i,E}^\top \hat{\theta}_n^E)\right] = 0,$$

since $\tilde{W}_{n,i}$ is scaled mean zero noise sampled independently from Y_i. Furthermore,

$$\frac{1}{n}\sum_{i=1}^{n} X_{i,E}\mathbb{E}\left[\tilde{W}_{n,i}^2\right] X_{i,E}^\top = \frac{1}{n}\sum_{i=1}^{n} X_{i,E}\,\mathrm{Var}\left(\tilde{W}_{n,i}\right) X_{i,E}^\top = \mathrm{Var}\left(\hat{W}_{n,E,E}\right),$$

since $\hat{W}_n := n^{-1/2}\sum_{i=1}^{n} X_i\tilde{W}_{n,i}$ as specified in Corollary 7. Thus,

$$\begin{aligned}
&\frac{1}{n}\sum_{i=1}^{n}\left(Y_i - \frac{1}{\gamma}\tilde{W}_{n,i} - \dot{b}(X_{i,E}^\top\hat{\theta}_n^E)\right)^2 X_{i,E}X_{i,E}^\top \\
&= \frac{1}{\gamma^2}\,\mathrm{Var}\left(\hat{W}_{n,E,E}\right) + \frac{1}{n}\sum_{i=1}^{n}\left(Y_i - \dot{b}(X_{i,E}^\top\hat{\theta}_n^E)\right)^2 X_{i,E}X_{i,E}^\top + o_p(1),
\end{aligned}$$

and so the the sandwich estimators in Algorithms 1 and 2 are equivalent up to an $o_p(1)$ term.

C.2 Additional simulation results

While data thinning is not applicable to Bernoulli data, it can be applied to binomial data [Neufeld et al., 2024, Dharamshi et al., 2025]. In the case of logistic regression with discrete covariates, it may be possible to group the Bernoulli observations to form binomial observations. Binomial data thinning of these binomial observations thus permits valid inference after the lasso in a logistic regression model.

In Figure 6, we revisit the logistic regression experiment in Figure 3. However, for each sample size n, we duplicate the same unique 50 feature vectors to allow the observations to be binned. For example, when $n = 200$, there are four observations for each of the 50 unique feature vectors. We see that binomial data thinning and sample splitting perform similarly, while score thinning yields slightly narrower confidence intervals and lower FDR. Meanwhile, the fully conditional approach of Huang et al. [2024] performs poorly at smaller sample sizes.

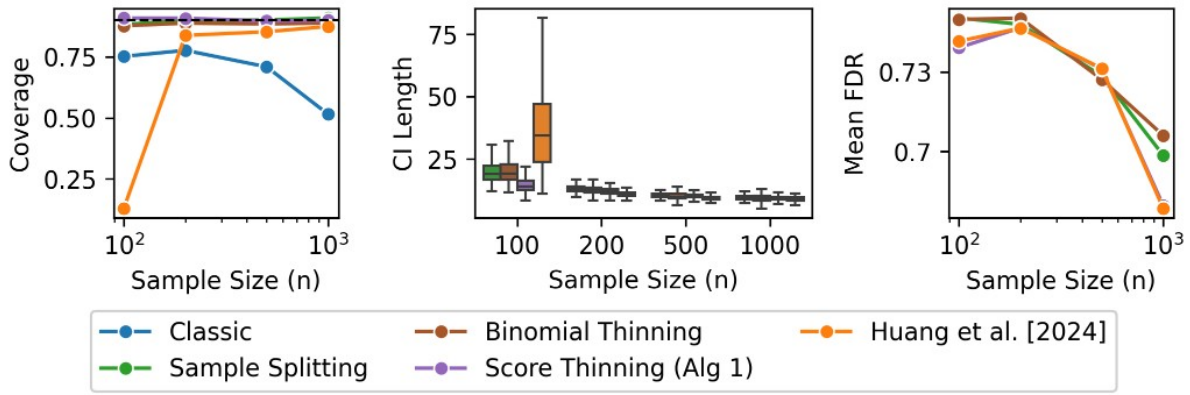


Figure 6: We simulate outcomes from a logistic regression model. There are 50 unique feature vectors, and so binomial thinning is applicable to groups of size $n/50$. Inference is conducted on the model selected using L_1-penalized logistic regression. Confidence intervals are computed for all coefficients in the selected model. Results are aggregated across 1000 simulated datasets. Since the classical approach (which reuses the data for selection and inference) fails to attain valid coverage, we omit it from the center and right-hand panels.